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Integrated single-cell RNA sequencing and Bulk-RNA technologies reveal the immunological characteristics of lactylation related-genes in glioblastoma

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Short Title:	Reveals the heterogeneity of macrophage lactylation in glioblastoma multiforme
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Keywords:	Glioblastoma, scRNA-seq, prognosis, Machine learning, Immunotherapy
Abstract:	Objective : Glioblastoma (GBM) stands as the most aggressive intracranial malignant tumor, characterized by an extremely poor prognosis. Lactylation, a recently identified post-translational modification, has been associated with tumorigenesis. However, its precise role in GBM is not yet fully understood. This study seeks to integrate single-cell RNA sequencing (scRNA-seq) and bulk RNA sequencing (RNA-seq) data to develop a novel prognostic model for GBM, concentrating on lactylation-related factors. Methods: We have studied lactate metabolism genes as a marker in GBM. We obtained bulk transcriptomic data from TCGA and the GSE141383 and GSE162631 cohorts from GEO databases. We used R package Seurat to analyze scRNA-seq data, cell communication using CellChat R package and AUCell package to assess the lactate metabolism gene set score across cell types. We developed a prognostic model using machine learning algorithms, and tested its efficacy across multiple cohorts. We also investigated differences between high and low risk groups in immune infiltration, predicted sensitivity, and other factors. We validated the function of the key gene CD93 on cell level. Results: The scRNA-seq data identified nine major cell types in GBM and FCGBP+ macrophages had the highest score of lactate metabolism gene set. Machine learning identified three key genes CD93, FCER1G and GRB2 and designed a model for optimal prognostic value across cohorts. The high-risk group had a high risk outcome. Immunological analysis showed significant differences in the immune cell infiltration and checkpoint gene expression between risk groups. High-risk patients had lower immune infiltration and higher immunosuppression, making them less suitable for immunotherapy. Predictive algorithms suggested that axitinib and imatinib could be potential therapeutic drugs for high-risk patients. In GBM tissue and cells, CD93 expression was significantly increased, thus CD93 was recognized as a key risk gene in this model. Inhibition of CD93 expression by siRNA significantly reduced the proliferation, invasion and migration of U87 and U251 cells. Conclusion:In summary, we developed a novel characterization of lactylation-related clusters using single-cell sequencing technology. This study offered insights into the prognostic significance of lactate metabolism-related genes in GBM.
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Response to Reviewers:	Reviewer #1: Line 46: "predicted sensitivity" may be prefered here.

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Response:
We appreciate the reviewer's careful suggestion. We have revised the relevant text at Line 46, replacing the original expression with "predicted sensitivity" as recommended. This revision makes the description more accurate and consistent with the research context.

- Line 52: Without an external *clinical* confirmation in large scale cohorts, "optimized" might be a preferable wording over "optimal", highlighting the process without being able to claim optimal performance in real world situations.
Response:
We appreciate the reviewer’s careful and constructive comment. Following the suggestion, we have revised the word “optimal” to “optimized” at Line 52. This change better emphasizes the model construction process rather than overstating its real-world performance without large-scale external clinical validation.

- Data processing, Line 130 and below: The authors claim that they filtered out cells with >10% mtRNA. However, in their response letter, they state 20%. Please check for accuracy. I also recommend stating the percentage of total variance captured by these 15 PCAs in the manuscript body and not only in the response letter.
Response:
We greatly appreciate the reviewer for catching this critical inconsistency and for the constructive suggestions.We have carefully checked and corrected the mitochondrial RNA filtering threshold in the manuscript (Line 130 and below). The threshold used in our analysis was 10% mtRNA, and we have revised the response letter accordingly to ensure full consistency across the manuscript and response materials.Following the reviewer’s recommendation, we have added the percentage of total variance captured by the 15 principal components into the main text of the manuscript, so that this key information is clearly reported and visible to readers.All corrections have been made in the revised manuscript.

- Ro/e analysis, Line 146 and below: In this paragraph, the authors spell "Role" and not "Ro/e" multiple times, which is somewhat confusing.
Response:
We thank the reviewer for pointing out this typographical error.We have carefully checked the paragraph at Line 146 and below, and corrected all misspelled “Role” back to “Ro/e” to avoid confusion and ensure consistency in the manuscript.

- Line 257: Please make sure that all reagents are denoted with their catalogue number and manufacturer. This can either be done in brackets after the first mention of a reagent or in a supplementary reagent table.

Response:
We appreciate the reviewers' constructive suggestions.
We thoroughly reviewed all the reagents listed in the manuscript and included the respective manufacturers in parentheses. The entire manuscript was meticulously checked multiple times to ensure that the source information for all reagents is clearly indicated.

- Line 334: The line still somewhat overstates the results of this study, "crucial". The same occurs in line 485.
Response:
We appreciate the reviewers' detailed comments. We agree that using "crucial" in Lines 334 and 485 overstates our study's conclusions. Consequently, we revised these sentences, replacing "crucial" with more suitable and moderate terms to accurately reflect our findings.

- Discussion: Multiple parapgraphs in the discussion are doubled, as if copied and pasted multiple times. This suggests that the authors have not proof-read the final manuscript. Please make sure that the revised manuscript is free of apparent mistakes. E.g., lines 463-466 & 476; 486 - 517 in multiple snippets.
Response:
Dear Reviewers,We sincerely apologize for the repeated paragraphs and inadvertent errors in the discussion section. These issues resulted from inadequate proofreading prior to submission.

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	We meticulously revised the discussion section, eliminating redundant content, including the repetitive paragraphs in lines 463–466, line 476, and lines 486–517. We reorganized the paragraphs to maintain logical consistency and thoroughly proofread the manuscript to correct any noticeable errors. The revised manuscript was carefully reviewed to improve its readability and academic quality.
Additional Information:	
Question	**Response**
Financial Disclosure Enter a financial disclosure statement that describes the sources of funding for the work included in this submission. Review the submission guidelines for detailed requirements. View published research articles from *PLOS ONE* for specific examples. This statement is required for submission and **will appear in the published article** if the submission is accepted. Please make sure it is accurate. **Funded studies** Enter a statement with the following details: • Initials of the authors who received each award • Grant numbers awarded to each author • The full name of each funder • URL of each funder website • Did the sponsors or funders play any role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript? Did you receive funding for this work?	The author(s) received no specific funding for this work.
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Data Availability Authors are required to make all data underlying the findings described fully available, without restriction, and from the time of publication. PLOS allows rare exceptions to address legal and ethical concerns. See the PLOS Data Policy and FAQ for detailed information. A Data Availability Statement describing where the data can be found is required at submission. Your answers to this question constitute the Data Availability Statement and **will be published in the article**, if accepted.	Yes - all data are fully available without restriction

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Additional data availability information:	Tick here if the URLs/accession numbers/DOIs will be available only after acceptance of the manuscript for publication so that we can ensure their inclusion before publication.
Ethics Statement Enter an ethics statement for this submission. This statement is required if the study involved: • Human participants • Human specimens or tissue • Vertebrate animals or cephalopods • Vertebrate embryos or tissues • Field research Write "N/A" if the submission does not require an ethics statement. General guidance is provided below. Consult the submission guidelines for detailed instructions. **Make sure that all information entered here is included in the Methods section of the manuscript.**	N/A

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Data Availability Before publication, Authors are required to make fully available and without restriction all data underlying their findings. Sharing the data underlying your findings increases the visibility, credibility, and impact of your work, while supporting transparency and future scientific discovery. Please see our <a href="https://journals.plos.org/plosone/s/data-availability" target="_blank">PLOS Data Policy page</a> for detailed information on this policy.</p> A Data Availability	The data that support the findings of this study are openly available in Gene Expression Omnibus (GEO) at https://www.ncbi.nlm.nih.gov/geo/, reference number GSE141383 , GSE162631.

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Statement, detailing where the data can be accessed, is required at first submission. Insert your Data Availability Statement in the box below.</p> Please see the <a href="https://journals.plos.org/plosone/s/submission-guidelines#loc-data-reporting" target="_blank">data reporting</a> of our submission guidelines for instructions on what you need to include in your Data Availability Statement. Examples of acceptable Data Availability Statements are included here for reference:</p> <ol> <li>The minimal data set [and accompanying code, where generated] is available at [repository name] via [DOI hyperlink]</li> <li>No data was generated by this study. The following existing data sources were used: [dataset name] from [dataset location] available via [web address or DOI hyperlink].</li> <li>For Study Protocols: No datasets were generated or analysed during the current study. All relevant data from this study will be made available upon study completion.</li> </ol> PLOS allows rare exemptions to address legal and ethical concerns. If you have legal or ethical restrictions, please use the box below to detail these in full sentences for the Journal team to consider.</p> This statement will be typeset if the manuscript is accepted for publication.</p>	

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Integrated single-cell RNA sequencing and Bulk-RNA technologies reveal the immunological characteristics of lactylation related-genes in glioblastoma

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Acknowledgements: None.

Abstract

Objective:

Glioblastoma (GBM) stands as the most aggressive intracranial malignant tumor, characterized by an extremely poor prognosis. Lactylation, a recently identified post-translational modification, has been associated with tumorigenesis. However, its precise role in GBM is not yet fully understood. This study seeks to integrate single-cell RNA sequencing (scRNA-seq) and bulk RNA sequencing (RNA-seq) data to develop a novel prognostic model for GBM, concentrating on lactylation-related factors.

Methods: We have studied lactate metabolism genes as a marker in GBM. We obtained bulk transcriptomic data from TCGA and the GSE141383 and GSE162631 cohorts from GEO databases. We used R package Seurat to analyze scRNA-seq data, cell communication using CellChat R package and AUCell package to assess the lactate metabolism gene set score across cell types. We developed a prognostic model using machine learning algorithms, and tested its efficacy across multiple cohorts. We also investigated differences between high and low risk groups in immune infiltration, predicted sensitivity, and other factors. We validated the function of the key gene CD93 on cell level.

Results: The scRNA-seq data identified nine major cell types in GBM and FCGBP+ macrophages had the highest score of lactate metabolism gene set. Machine learning identified three key genes CD93, FCER1G and GRB2 and designed a model for optimal prognostic value across cohorts. The high-risk group had a high risk outcome. Immunological analysis showed significant differences in the immune cell infiltration and checkpoint gene expression between risk groups. High-risk patients had lower immune infiltration and higher immunosuppression, making them less suitable for immunotherapy. Predictive algorithms suggested that axitinib and imatinib could be potential therapeutic drugs for high-risk patients. In GBM tissue and cells, CD93 expression was significantly increased, thus CD93 was recognized as a key risk gene in this model. Inhibition of CD93 expression by siRNA significantly reduced the proliferation, invasion and migration of U87 and U251 cells.

Conclusion: In summary, we developed a novel characterization of lactylation-related clusters using single-cell sequencing technology. This study offered insights into the prognostic significance of lactate metabolism-related genes in GBM.

Keywords:Glioblastoma, scRNA-seq, prognosis, Machine learning, Immunotherapy.

Introduction

GBM is one of the most common primary intracranial tumors [1-2]. It is the most aggressive tumor of brain neuroglia and is the most commonly encountered primary

intracranial tumor [3-5]. It is rapidly growing and densely infiltrating the brain and hence the deadliest tumor of the central nervous system [3-5]. Initial symptoms include nonspecific cognitive deficits and seizures [5]. Surgery, radiotherapy, and chemotherapy drugs are standard treatments. The median survival rate for GBM patients is under one year [6]. Recent work has identified several biomarkers that provide valuable information about the molecular subtypes of GBM, their prognosis and treatment responses [7]. However, due to its rapid growth and heterogeneous nature, it still remains difficult to treat for oncology. Finding therapeutic targets and developing prognostic risk models are critical for optimal clinical management of glioblastoma.

Since Warburg effect was introduced in 20th century, research has shown lactic acid plays a key role in initiation and progression of malignant tumors [8].Glucose is considered to be the main energy source for tumor cells but tumor cells need a large amount of nutrients and oxygen for rapidly expanding tumor cells [9]. According to Warburg effect, tumor energy metabolism depends more on anaerobic glycolysis than oxidative phosphorylation even in the presence of oxygen. Increased lactic acid levels in tumor microenvironment lead to hypoxic microenvironment [10]. High levels in tumor microenvironment correlate to tumor growth, metastasis, and tumor cell evasion [11]. Tumor cells export large amounts of lactic acid to adjacent immune and stromal cells. Thus, lactic acid acts as mediator between endogenous metabolic processes and immune suppression [12]. She X[13] and colleagues showed that SETDB1 methylation inhibits interaction between MCT1 and Tollip and induces M2-like polarization of tumor-associated macrophages. Su J[14] and colleagues found that absence of lactic acid/GPR81 inhibits expression of chemokine CX3CL1, decreases Treg infiltration into tumor microenvironment (TME) and inhibits gastric cancer progression by improving CD8+ T cells function. The mechanisms of lactic acid production pathways or its transport to lower lactic acid concentrations is not fully understood.

We used scRNA-seq to investigate the role of lactylation in GBM cell diversity. We developed and validated a novel model using signature genes from FCGBP+

macrophages, which showed highest lactylation scores. We also studied the relationship between risk model and immune-infiltrating cells to identify potential therapeutic targets. We identified CD93 as important gene in the model and studied its effects on GBM cell function at cell level.

Methods

Data Collection

The data in this article were sourced from publicly accessible databases, specifically the TCGA (https://portal.gdc.cancer.gov/) and GEO (https://www.ncbi.nlm.nih.gov/geo/). The single-cell datasets used in this study, GSE141383 and GSE162631, include 13 GBM tissue samples and 4 control tissue samples. The bulk datasets are divided into two parts: 1) The training dataset for the diagnostic model comes from the TCGA-GBM cohort, comprising 5 control tissue samples and 168 GBM tissue samples, with survival information for 80 GBM patients; 2) The validation dataset for the diagnostic model is obtained from the CGGA-GBM database, which includes 137 GBM patients with survival information, and the GEO database (GSE83300), which contains 50 GBM patients with matched survival data.

Data Processing

For single-cell data analysis, Seurat version 4.2.2 was employed. Initially, quality control excluded cells with fewer than 200 or more than 5000 nFeature_RNA and those with a mitochondrial proportion over 10%. Data normalization and standardization were conducted using the SCTransform function. Dimensionality reduction and clustering were performed by selecting the top 15 principal components, and the RunUMAP function was used to generate results. The DoubletFinder tool removed doublets from the dataset. To address batch effects across datasets and samples, the Harmony algorithm was applied. Cell annotation was conducted using the SingleR method alongside existing literature. The FindAllMarkers function identified marker genes for each cell type. UMAP results were visualized using the ggplot2 package, and cell proportions were represented with the ggalluvial package.

For bulk RNA data, normalization and standardization were performed using the limma package. Additionally, survival analysis and curve plotting were executed with the Survival and survminer packages.

Ro/e Analysis

The Ro/e ratio [15] illustrates the relationship between observed and expected cell counts for each cell cluster across different tissue types. Expected cell counts for each cell cluster and tissue type combination are derived from chi-square tests. Unlike chi-square values, which only indicate the difference between observed outcomes and random expectations, Ro/e values reveal whether specific cell clusters are enriched or depleted in particular tissues. A Ro/e value greater than 1 suggests that a cell cluster is more prevalent in a specific tissue than expected by chance, indicating enrichment. Conversely, a Ro/e value less than 1 indicates that a cell cluster is less prevalent than expected, suggesting depletion. By employing Role calculations, researchers can effectively evaluate the tissue preferences of various cell clusters.

Cell Communication

The "CellChat" R package (version 1.5.0) is a tool designed to elucidate potential mechanisms of intercellular communication at the single-cell level. The function createCellChat generates a cellchat object, while the aggregateNet feature outlines the signaling outputs for each cell type. The netAnalysis_computeCentrality function calculates the input and output weights of specific signaling pathways. Data visualization is supported by the "CCPlotR" package, where the cc_heatmap function shows the frequency of intercellular communications, and the cc_arrow function illustrates receptor-ligand interactions relevant to target cells.

lactylation geneset score

Gene sets related to lactate were obtained from the MSigDB database (https://www.gsea-msigdb.org/). We calculated the lactylation gene set score using the AUCell package, specifically the AUCell_calcAUC function. This approach enabled

us to compare lactate metabolism gene set scores across various cell types, including macrophage subclusters identified through secondary clustering, to identify cell subpopulations associated with lactate. We visualized the lactate metabolism gene set scores using ggplot2, presenting them as violin plots.

Machine Learning and Prognostic Model Construction

Marker genes linked to the primary lactate-producing cell subtypes are identified using criteria of a log fold change (logFC) greater than 0.5 and a p-value less than 0.001. Core gene identification is performed using three machine learning methods: 1) LASSO regression with the "glmnet" package, 2) the CoxBoost algorithm via the "CoxBoost" package with an initial penalty of 1000, and 3) a Random Forest survival model through the "RandomForestRSF" package, using 1000 trees and evaluating gene significance with importance scores. All methods incorporate cross-validation. A Venn diagram analyzes the intersection of core genes identified by these algorithms to determine core risk prognostic genes. A multivariate regression analysis then develops the prognostic model. Risk scores are calculated by summing gene expression values weighted by their risk coefficients. GBM patients are classified into high-risk and low-risk groups based on the median risk score. Survival analyses and survival curves are generated using the "Survival" and "survminer" packages, while Receiver Operating Characteristic (ROC) curves are created with the "timeROC" package. The "ggplot2" package is used to produce Principal Component Analysis (PCA) plots, aiding in identifying key characteristics of the risk groups and enabling dimensionality reduction clustering.

Enrichment Analysis

Enrichment analysis includes Gene Ontology (GO), Kyoto Encyclopedia of Genes and Genomes (KEGG), and HALLMARK enrichment analyses. The clusterProfiler package is used for Gene Set Enrichment Analysis (GSEA) across these analyses, applying a threshold of an adjusted p-value below 0.05. For visualizing GO enrichment, the aPEAR package is used, while ggplot2 is applied for

188 KEGG enrichment visualization. Additionally, the GseaVis package is utilized to
189 visualize HALLMARK enrichment analysis.
190
191 **Immunology-related Analysis**
192 ssGSEA is a widely used method for evaluating immune cell types, focusing on
193 29 distinct categories and related immune processes. The "GSVA" and "GSEABase"
194 packages calculate relative scores for these immune cells and processes. For
195 visualization, the "ggpubr" and "introdataviz" packages create split violin plots. The
196 "Estimate" package analyzes the tumor microenvironment, and correlation analyses
197 explore relationships between risk scores and immune, stromal, and
198 microenvironment scores. The "TCGAbiolinks" package retrieves single nucleotide
199 polymorphism (SNP) data for GBM patients, while "maftools" calculates tumor
200 mutational burden (TMB) scores for each patient. Additionally, the TIDE website
201 (http://tide.dfci.harvard.edu/) assesses immune evasion and dysfunction status. Finally,
202 "ggpubr" generates boxplots to depict the expression levels of immune checkpoint
203 molecules.
204 **Drug Sensitivity Analysis**
205 The pRRophetic algorithm was used to calculate drug sensitivity IC50 values,
206 aiding in predicting patient responses to chemotherapeutic and targeted agents across
207 different cohorts. By analyzing the correlation between risk scores and drug IC50
208 values, drugs with positive correlation coefficients emerged as potential therapeutic
209 options for low-risk groups. In contrast, drugs showing a significant negative
210 correlation with IC50 values were identified as potentially suitable for high-risk
211 patients.
212
213 **Cell Culture**
214 Human brain astrocytes SVG p12 (FH0563) and glioblastoma cell lines U251
215 (FH0159) and U87 (FH0162) were obtained from Fuheng Biological Technology Co.,
216 Ltd. (Shanghai, China). These cells were cultured in DMEM medium (Gibco, Grand
217 Island, NY, USA) supplemented with 10% fetal bovine serum (FBS; Gibco), 100

U/mL penicillin, and 100 μg/mL streptomycin (Invitrogen, Waltham, MA, USA). Cultivation occurred in a humidified incubator set at 37°C with 5% CO2.

Cell Transfection

The siRNA targeting CD93 was designed and synthesized by Tsingke Biotechnology Co., Ltd. (Beijing, China). Transfection was performed using Lipofectamine 3000 reagent (Invitrogen, Waltham, MA, USA) according to the manufacturer's instructions. Post-transfection, cells were cultured at 37°C with 5% CO2 for 48 hours. The study comprised three experimental groups: cells transfected with si-CD93#1, si-CD93#2, and a negative control group (si-NC). The specific siRNA sequences used in the study were as follows:

si-CD93#1:GGCUACUGGUCUAUCGCAA;

si-CD93#2: GUGCAAGUUCAGCUUCAAA;

Cell Viability Assay

Cell viability was assessed using the CCK-8 assay kit (Invigentech, IV08, USA). After 48 hours of siRNA transfection, U251 and U87 cell lines were seeded into 96-well plates at a density of 3,000 cells per well. At 24, 48, and 72 hours post-transfection, 10 μL of CCK-8 reagent was added to each well, followed by a 2-hour incubation at 37°C. The optical density (OD) at 450 nm was then measured with a microplate reader to determine the relative cell survival rate.

Transwell Invasion Assay

The invasion assay employed Transwell chambers with 8μm pores (Corning, USA, Cat. No. 3422) placed in a 24-well plate. The upper chamber was coated with an 8-fold dilution of BD Matrigel. U251 and U87 cell lines, transfected with siRNA for 48 hours, were digested with trypsin and resuspended at 3×104^5 cells per chamber in 200μL of serum-free medium. These cells were added to the upper chamber, while 600μL of medium containing 20% fetal bovine serum (FBS) was placed in the lower chamber. The setup was incubated at 37°C with 5% CO2 for 48

248 hours. After incubation, non-migratory cells were removed from the upper chamber,
249 and cells adhering to the membrane underside were fixed with 4% paraformaldehyde
250 for 20 minutes. Following fixation, the cells were stained with crystal violet. Cell
251 invasion was observed and documented using a microscope..
252
253 **Scratch assay**
254 Forty-eight hours after siRNA transfection, U251 and U87 cell lines were
255 re-plated into 6-well plates and cultured until they reached about 90% confluence
256 within 24 hours. Using sterile 10 μl pipette tips, three linear scratches were made on
257 the cell monolayer. The wells were then rinsed with phosphate-buffered saline (PBS)
258 to remove any detached cells. A medium containing 4% fetal bovine serum (FBS) was
259 added to each well, and the cells were further cultured. The wound healing process
260 was observed, and images of the scratched regions were captured at 0 and 24 hours
261 using a microscope.
262
263 **Quantitative Real-Time PCR (qRT-PCR)**
264 U251 and U87 cells were collected 48 hours post-siRNA transfection. The
265 procedure followed the kit's instructions. Total RNA was extracted using the RNAiso
266 Plus Reagent Kit (Takara, 9108/9109, Japan). mRNA was then reverse-transcribed
267 into cDNA with the PrimeScript RT Reagent Kit (Takara, RR037A, Japan). Real-time
268 quantitative PCR was conducted using the TB Green Premix Ex Taq II Reagent Kit
269 (Takara, RR820A, Japan). GAPDH served as the normalization control, and CD93
270 mRNA expression levels were determined using the 2-ΔΔCt method. The primer
271 sequences are as follows:
272 CD93-F: 5'-GGCAGACAGTTACTCCTGGGTT-3',
273 CD93-R: 5'-GGAGTTCAAAGCTCTGAGGATGG-3';
274 GAPDH-F：5’- ACAGCCTCAAGATCATCAGCA-3’;
275 GAPHD-R: 5’-ATGAGTCCTTCCACGATACCA-3’;
276
277 **Data Analysis**

Statistical analyses utilized R software (version 4.3.1). Continuous variables were analyzed with either Student's t-test or the Mann-Whitney U test, while categorical variables were assessed using the chi-square test or Fisher's exact test. Kaplan-Meier survival analysis estimated survival outcomes. For analyses with multiple comparisons, such as differences in clinical features across risk groups, P-values were adjusted for the false discovery rate using the Benjamini-Hochberg method. Statistical significance was denoted as follows: *P < 0.05, **P < 0.01, ***P < 0.001, and NS for no significant difference.

3 Result

The scRNA profiling of GBM

To thoroughly understand the cell types and molecular changes linked to GBM, we integrated two single-cell datasets (GSE141383 and GSE162631) and used the Harmony algorithm to correct for batch effects. After implementing stringent control measures, we analyzed 114,630 cells, including 45,282 control cells and 69,348 primary GBM cells. Using a resolution parameter, we categorized these cells into nine distinct populations (Fig. 1A). Specific annotations were based on markers for macrophages (OC1, SPP1, CD13), microglia (CX3CR1, P2RY12, P2RY13), neuron-like cells (PTN, PTPRZ, BCAN), monocytes (VCAN, LYZ, THBS1), neutrophils (IL12, CSF3R), T cells (CD2, CD3D, CD3E), endothelial cells (CLDN5, VWF), fibroblasts (RGS5, DCN, COLA1), and proliferating cells (MKI67, TOP2A, TPX2) (Fig. 1B).To identify cell types closely associated with GBM progression, we conducted an observed-to-expected ratio (Ro/e) analysis, a method commonly used to assess tissue preference. Results indicated that microglia and endothelial cells were predominantly enriched in control tissues, while other cell types were concentrated in GBM tissues (Fig. 1C). This finding suggests that tumorigenesis mobilizes various cells to collaboratively reshape the tumor microenvironment. We further evaluated the relative differences in cell types between GBM and control tissues. Notably, macrophages, neuron-like cells, monocytes, and neutrophils significantly increased in

GBM, highlighting their critical roles in GBM development and progression (Fig. 1D).Cell communication analysis revealed that macrophages and microglia exhibited the highest incoming and outgoing signal intensities (Fig. 1E). Although increasing evidence suggests that excessive lactylation activation is closely associated with tumor progression, its impact on GBM progression remains unclear. We obtained a lactylation-related gene set from the MSigDB database and applied the AUCell score to evaluate the nine cell types. Macrophages had the highest lactylation gene set score (Fig. 1F), leading us to speculate that macrophages may promote GBM progression through the lactylation pathway.Subsequently, we performed a secondary clustering analysis of macrophages and identified 11 different macrophage subtypes (Fig. 1G). Interestingly, the FCGBP+ macrophage subtype showed the highest lactylation gene set score (Fig. 1H). Additionally, we compared lactylation gene set scores between the control and GBM groups, finding that the GBM group had a significantly higher score ($P < 0.01$), further confirming lactylation as a important factor in GBM occurrence and development (Fig. 1I).

Machine Learning Construct FCGBP+ Macrophages Signatures

We identified marker genes of FCGBP+ macrophages using screening criteria (logFC > 0.5, P 0.001) and validated their expression in control and GBM tissues of TCGA-GBM cohort. We found 12 genes upregulated and 9 downregulated in GBM (Fig. 2A). GO enrichment analyses of these differentially expressed genes showed their role in immune cell regulation, cell differentiation, and other biological processes (Fig. 2B). To identify core genes of FCGBP+ macrophages, we applied three machine learning algorithms. CoxBoost algorithm obtained the most stable model at the 117th boost and identified 3 core genes with partial log log 210.4623 (Fig. S1A); LASSO regression allowed shrinkage and cross validation and retained 3 most reliable core genes lambda.min = 0.1060313 (Fig. S1B-C). Random forest survival model obtained the lowest error rate of 0.4783 at 143 trees, ranking differential genes by importance and identified 4 genes with an importance score above zero (Fig. S1D-E).We constructed Venn diagram from outputs of the three

337 machine learning algorithms and selected intersecting genes CD93, FCER1G, and
338 GRB2 as core genes (Fig. 2C).
339
340 **Validation of Prognostic Features and Clinical Value**
341 We developed a risk model based on three lactylation genes: CD93, FCER1G,
342 and GRB2. We calculated the risk score for each GBM patient using Lasso analysis
343 and multivariate Cox regression analysis. Patients were then classified into high-risk
344 and low-risk groups based on the median risk score. In the training cohort
345 (TCGA-GBM dataset) and two validation sets (CGGA dataset, GSE83300), we
346 observed a significant difference in overall survival between these groups. High-risk
347 patients exhibited lower survival rates compared to low-risk patients ($P < 0.05$) (Fig.
348 3A-C).To evaluate the predictive power of these three genes for GBM prognosis, we
349 conducted time-dependent receiver operating characteristic (ROC) curve analyses in
350 both training and validation sets. The average area under the curve (AUC) of the risk
351 model at 1, 2, and 3 years was 0.55, highlighting its biological significance (Fig.
352 3D-F). We also identified significant biological processes and pathways
353 differentiating low-risk and high-risk patients. Gene Ontology (GO) enrichment
354 analysis revealed that low-risk patients had higher immune effector processes and
355 increased activation and expression of mitochondrial genes (Fig. 3G).Further, KEGG
356 enrichment analysis showed that pathways related to cytokine receptor interaction,
357 antigen presentation, and oxidative phosphorylation were upregulated in the low-risk
358 group, while cancer pathways such as WNT and NOTCH were downregulated (Fig.
359 3H). These findings help elucidate the reasons for the differences observed between
360 the risk groups. The low-risk group had a higher mutation frequency (94.59%), with
361 TP53 being the most commonly mutated gene (Fig. 3I-J). In contrast, the high-risk
362 group had a mutation frequency of 86.84%, with mutations primarily in the PTEN
363 gene (Fig. 3I-J).
364
365 **Immune-Related Features of the Risk Model**
366 We assessed immune infiltration across various risk categories and core genes

using the ssGSEA algorithm to calculate scores for 29 distinct immune cell types and processes. Our findings indicated that the low-risk group had higher infiltration levels of dendritic cells (DCs), neutrophils, T helper cells, and tumor-infiltrating lymphocytes (TILs). This group also exhibited enhanced T cell costimulation and type I interferon responses (Supplementary Fig. S2A). Additionally, the core genes associated with the three models showed unique relationships with TNF family genes and chemokines. Notably, FCER1G had significant negative correlations with most TNF family genes and chemokines, while CD93 and GRB2 showed partial positive correlations with these genes and chemokines (Supplementary Fig. S2B-C). To evaluate TME activity, we used TME scores and discovered that high-risk scores were significantly negatively correlated with both immune scores and TME scores. This suggests that lower risk scores may be associated with more immunologically active microenvironments and relatively lower tumor purity (Supplementary Fig. S2D-F).

Immunotherapy Efficacy Stratification by Immune Microenvironment Risk Model

We evaluated the effectiveness of immunotherapy across different risk categories and discovered that the tumor mutation burden (TMB) was higher in the low-risk cohort. Furthermore, we found a significant negative correlation between microsatellite instability (MSI) and risk scores, indicating that low-risk patients have elevated levels of both TMB and MSI, making them more suitable for immunotherapy (Fig. 4A-B). Conversely, the high-risk group showed significantly increased TIDE scores and a higher prevalence of tumor-associated fibroblasts and myeloid-derived suppressor cells (MDSCs), suggesting a connection between high-risk status and less favorable immunotherapy outcomes (Fig. 4C-F). Our analysis also revealed that immune checkpoint molecules were predominantly expressed at higher levels in the low-risk group, reinforcing the notion that these patients are more likely to benefit from immunotherapy (Fig. 4G). Therefore, individuals in the low-risk group may represent a promising demographic for immunotherapeutic strategies.

397 **Analysis of Antitumor Drugs in High-Risk and Low-Risk Groups**

398 We conducted a drug sensitivity prediction analysis for patients across various
399 risk groups, utilizing the pRRophetic algorithm to determine IC50 values. Axitinib
400 and Imatinib displayed a significant negative correlation with the risk score,
401 indicating their potential effectiveness for high-risk patients (Fig. 5A-B). Conversely,
402 seven drugs, including AZD6244, showed a positive correlation with the risk score,
403 suggesting their appropriateness for low-risk patients. Notably, metformin exhibited
404 the strongest and statistically significant correlation (Fig. 5C-I).

405

406 **CD93 is potentially a important gene in models linked to poor prognosis in GBM**

407 To identify key molecular biomarkers for our prognostic model, we evaluated the
408 expression levels of three critical genes in GBM. Our analysis revealed that only
409 CD93 and FCER1G were significantly upregulated in GBM tissues, suggesting their
410 potential roles in tumor progression (Fig. 6A). Kaplan-Meier survival analysis further
411 indicated that CD93 was the only gene with a statistically significant impact ($P<0.05$)
412 on GBM patient survival, with higher CD93 expression associated with poorer overall
413 survival outcomes (Fig. 6B-C). Therefore, CD93 emerged as a promising molecular
414 biomarker for GBM.Interestingly, CD93 did not significantly affect the tumor antigen
415 presentation process but showed a positive correlation with HLA family molecules
416 (Fig. 6D), implying involvement in alternative tumor-promoting mechanisms.
417 Additionally, hallmark enrichment analysis compared pathway variations between
418 groups with high and low CD93 expression. Patients with elevated CD93 expression
419 demonstrated increased activity in the epithelial-mesenchymal transition (EMT),
420 interleukin-6 (IL6), and KRAS pathways. In contrast, those with lower CD93
421 expression exhibited upregulation of processes such as oxidative phosphorylation (Fig.
422 6E-J). These pathways may elucidate how CD93 contributes to GBM progression.

423

424 **CD93 is upregulated in GBM cell lines and enhances GBM cell proliferation,**
425 **invasion, and migration**

426 In summary, the findings indicate that CD93 is a significant gene associated with

poor prognosis in GBM. Quantitative reverse transcription polymerase chain reaction (qRT-PCR) analyses showed that human astrocytoma cells had higher CD93 expression levels compared to human brain astrocytes (Fig. 7A). We performed simultaneous transfections of siRNA#1 and #2 into U251 and U87 cell lines, which confirmed the transfection efficiency of CD93 (Fig. 7B-C). Cell Counting Kit-8 (CCK8) assays demonstrated that knocking down CD93 significantly inhibited the proliferation of both U251 and U87 cells (Fig. 7D). Additionally, scratch and Transwell assays revealed that CD93 knockdown decreased the migratory and invasive abilities of U251 and U87 cells. Overall, these results suggest that CD93 acts as a positive regulator in the context of GBM.

Discussion

This study innovatively integrates single-cell sequencing data with extensive transcriptome datasets from public databases to investigate the lactic acid metabolism mechanism in macrophages within the GBM tumor microenvironment. We performed a comprehensive analysis of GBM by incorporating large-scale transcriptome data. Utilizing single-cell sequencing, machine learning, and immune infiltration analysis, we examined the effects of lactation modification on GBM. Our findings suggest that genes associated with lactation modification could serve as novel therapeutic targets. Additionally, we established a reliable patient survival prognosis model.The importance of lactation modification in the glioblastoma tumor microenvironment is increasingly acknowledged, with lactate dehydrogenase (LDH) identified as a major prognostic marker for this malignant tumor [16]. Tumor cells often undergo metabolic reprogramming, resulting in excessive lactic acid production, which promotes angiogenesis, immune evasion, and drug resistance [17-18]. Research by Yan et al. [19] confirmed that lactic acid enhances GPR65 expression on tumor-associated macrophages (TAMs), activating the cAMP/PKA/CREB signaling pathway, thereby facilitating glioblastoma invasion and metastasis. Consequently, lactic acid modification presents a promising target for GBM treatment.Further research is

essential to pinpoint specific targets related to lacylation, which could become new therapeutic targets for GBM. Developing innovative therapeutic strategies based on this metabolic pathway is prominent.

We obtained RNA expression profile data for GBM from the TCGA database, while gene mutation data and clinical information came from the external validation cohort in the GEO and CGGA databases. This comprehensive dataset enabled a thorough analysis of GBM's molecular characteristics. Additionally, we used single-cell RNA sequencing data to explore GBM heterogeneity within the tumor microenvironment, identifying nine distinct cell types: macrophages, microglia, neuron-like cells, monocytes, neutrophils, T cells, endothelial cells, fibroblasts, and proliferation-like cells. These findings provide meaningful insights into the mechanisms underlying GBM occurrence.We employed the lactic acid metabolism gene set score to identify individual cells involved in lactic acid metabolism. Macrophages exhibited the highest gene set score, aligning with Yang et al.'s findings [20], which reported a significant link between lactic acid metabolism and gastric cancer progression, with macrophages showing the highest infiltration levels. This supports our results. To pinpoint the primary cellular subtypes influencing lactic acid metabolism, we conducted secondary dimensionality reduction clustering, classifying macrophages into 11 subtypes. The FCGBP+ macrophage subtype displayed the highest lactate metabolism gene set score.Subsequently, we analyzed core genes related to the FCGBP+ macrophage subtype in the TCGA-GBM cohort, identifying 12 up-regulated and 9 down-regulated genes in GBM. These differentially expressed genes are involved in immune cell regulation, cell differentiation, and various biological pathways. Our research deepens the understanding of lactic acid metabolic activity variations in GBM cells.The primary objective of this study was to establish a robust prognostic model for GBM. Using machine learning algorithms, we constructed a predictive model based on key lactic acid metabolism-related genes. This model effectively stratifies patients into high-risk and low-risk groups. Survival analysis revealed that high-risk patients have shorter prognoses and survival times. Notably, the risk model, validated across the TCGA training set, CGGA validation set,

and GEO validation set, achieved an average area under the curve (AUC) of over 0.6 at 1, 2, and 3 years. This demonstrates the model's stability and accuracy in multi-cohort studies and its potential clinical application.Furthermore, our study conducted an in-depth analysis of the immune characteristics of GBM, revealing significant differences between high-risk and low-risk groups in immune cell infiltration, immune checkpoint gene expression, and tumor mutation burden. These findings suggest that the prognostic model's risk stratification mechanism may reflect immunological differences among GBM tumors in various populations, highlighting the need to explore differential immunotherapy responses among patient groups.

Through the analysis of core lactate-promoting genes, we identified CD93 as a key gene linked to poor prognosis in GBM. Polymeric protein-2 (MMRN2), a widely expressed extracellular matrix protein, specifically interacts with CD93, promoting endothelial cell adhesion and migration, which facilitates angiogenesis [21]. Additionally, CD93 is significantly upregulated in breast cancer, where its interaction with MMRN2 activates integrin β1, promoting tumor growth and angiogenesis via the PI3K/AKT/SP2 signaling pathway [22]. Thus, we speculate that CD93 could serve as a potential biomarker for GBM. Cell experiments confirmed that high CD93 expression in GBM cells, and inhibiting CD93 expression significantly reduced the proliferation, invasion, and migration abilities of U87 and U251 cells. This finding offers promising therapeutic targets for future GBM treatments.In conclusion, this study demonstrated that using the lactic acid fermentation gene set score as an indicator in single-cell datasets effectively identifies active cell clusters involved in lactic acid metabolism. Research found that this specific cell cluster is associated with a more aggressive disease phenotype, suggesting its important role in GBM progression. By integrating single-cell RNA sequencing data with bioinformatics technology, we identified key genes related to lactic acid fermentation and developed an innovative diagnostic model for predicting GBM. These results reveal lactic acid-mediated metabolic reprogramming and immune escape mechanisms in GBM, while identifying potential biomarkers and therapeutic targets. Further validation through larger patient cohorts is necessary, and the potential efficacy of targeting

lactate metabolic pathways in GBM treatment should be evaluated.

GBM:Glioblastoma;

TNF:Tumor Necrosis Factor;

TME:Tumor Microenvironment;

TAM:Tumor-Associated Macrophage;

CAF:Cancer-Associated Fibroblast;

EC:Endothelial Cell;

scRNA-seq:Single-cell RNA Sequencing;

GEO:Gene Expression Omnibus database;

TCGA:The Cancer Genome Atlas database;

AUC: Area Under the Curve

ROC： Receiver Operating Characteristic

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None

Competing interests

The authors declare that there are no competing interests.

Author Contributions

Biao Wang: experimental work, Conceptualization, Investigation, Writing–original draft. **Yangfang An**: Formal analysis. **Yuansen Shu**: Formal analysis,Writing–original draft. **Xiaoping Cheng**: Writing–review & editing.**Xuxiang Chen**: Writing – review & editing. **Xiezhuo Zhang**: Supervision, Project administration, Methodology. **Zhaorui Cheng**: Supervision, Project administration, Methodology.

Data availability statement

The data that support the findings of this study are openly available in Gene Expression Omnibus (GEO) at https://www.ncbi.nlm.nih.gov/geo/, reference number GSE141383，GSE162631.

Ethics aproval and consent to participate

This article does not contain any studies with human participants or animals.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

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Figures legends

Fig.1 Immune landscape of single-cell data in GBM

(A) Identification of nine cell types in GBM. (B) Gene annotation for these nine cell types. (C) Distribution comparison of the nine cell types between control and GBM tissues. (D) R o/e scores for each of the nine cell types. (E) Heatmap illustrating communication among the nine cell types. (F) Lactylation gene set scores for the nine cell types. (G) Comparison of lactylation gene set scores between normal and GBM tissues. (C) UMAP plot depicting 11 cell subtypes following secondary clustering of macrophages. (E) Presentation of lactylation gene set scores between control and GBM tissues. (F) Lactylation gene set scores across 11 macrophage subtypes. (G) Identification of 11 macrophage subtypes in GBM. (H) Comparison of lactylation gene set scores between normal and GBM tissues. (I) Lactylation gene set scores among the 11 macrophage subtypes.

Fig.2 Construction of FCGBP+ Macrophages Prognostic Genes

(A) The volcano plot illustrates the differentially expressed genes in FCGBP+ macrophages. (B) The GO enrichment analysis highlights the differentially expressed genes in FCGBP+ macrophages. (C) The Venn diagram shows the gene intersections identified by three distinct machine learning algorithms.

Fig.3 Validation and enrichment analysis of the risk prognostic model

(A-C) Kaplan-Meier survival curves illustrate overall survival (OS) for patients across various risk models within the TCGA, CGGA, and GSE83300 cohorts. (D-F) ROC curves depict 1-year, 2-year, and 3-year survival rates for different risk models in the

TCGA, CGGA, and GEO GSE83300 datasets. (G) GSEA pathway analysis compares the high-risk and low-risk groups. (H) KEGG pathway analysis contrasts the high-risk and low-risk groups. (I-J) mutational landscape plot provide a summary of mutation rates in high-risk versus low-risk populations. (*p<0.05; **p<0.01; ***p<0.001; NS: not statistically significant)

Fig.4 Characteristics of Multiple Immune Microenvironment Scoring Systems in Risk Models

(A)Analysis of Immune Mutation Burden in High-Risk and Low-Risk Groups (B)Correlation Analysis of MSI Scores Between High-Risk and Low-Risk Groups.(C)TIDE Score Analysis in High-Risk and Low-Risk Groups (D) CAF Score Analysis in High-Risk and Low-Risk Groups (E) MDSC Score Analysis in High-Risk and Low-Risk Groups.(F)TAM_M2 Score Analysis in High-Risk and Low-Risk Groups.(G)Correlation Analysis of Immune Checkpoint-Related Molecules Between High-Risk and Low-Risk Groups.*p<0.05; ** p<0.01; *** p<0.001;NS: not statistically significant.

Fig.5 Predicted drug sensitivity stratified by immune microenvironment risk score

（A–I）The pRRophetic algorithm predicts correlations between nine clinically relevant antitumor drugs and drug sensitivity in patients with varying risk scores.(A) Axitinib, (B) Imatinib, (C) AZD6244, (D) Bosutinib, (E) Camptothecin, (F) Erlotinib, (G) Lapatinib, (H) Metformin, (I) Sunitinib. Box plots compare the distribution of predicted sensitivity scores between low-risk and high-risk groups, where *p<0.05; ** p<0.01; *** p<0.001;NS: not statistically significant.

Fig.6 Prognostic model core gene screening

(A) Expression levels of CD93, FCER1G, and GRB2 in the TCGA-GBM cohort. (B-C) Survival curve analysis of CD93 and CER1G.(D) Correlation analysis of CD93 with human leukocyte antigen (HLA) family genes.(E-J)KEGG pathway enrichment

analysis of CD93-related genes. (*p<0.05; ** p<0.01; *** p<0.001;NS: not statistically significant).

Fig.7 Validation of the role of core gene CD93 in cytology experiments

(A) Expression levels of CD93 mRNA in GBM cell lines. (B-C) Knockdown efficiency of CD93 mRNA in U251 and U87 cells. (D) CCK8 analysis showed that interfering with CD93 expression significantly inhibited the proliferation ability of U251 and U87 cells.(E-F) Interfering with CD93 expression significantly inhibited the migration and invasion abilities of U251 and U87 cells. (G) Statistical graphs of invasion and migration experiments (*p<0.05, **p<0.01, ***p<0.001, ns: not significant).

Figure.1

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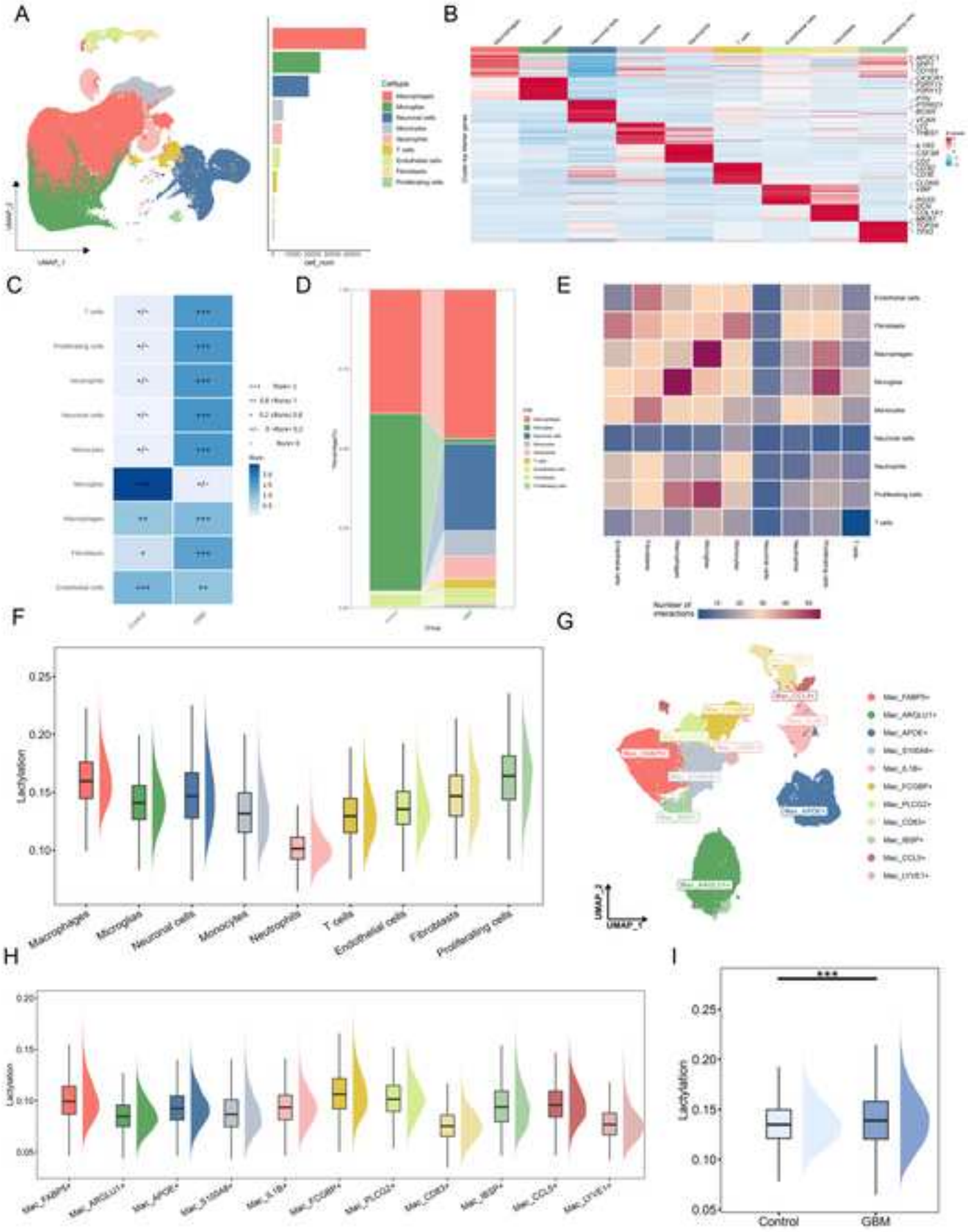

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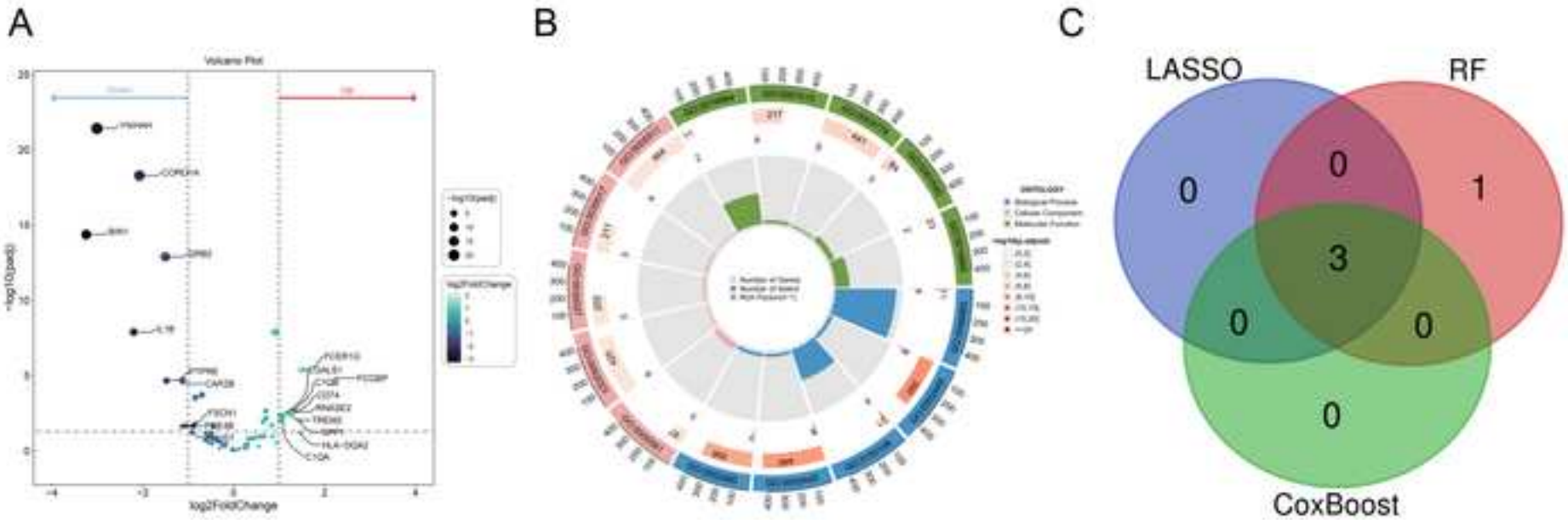

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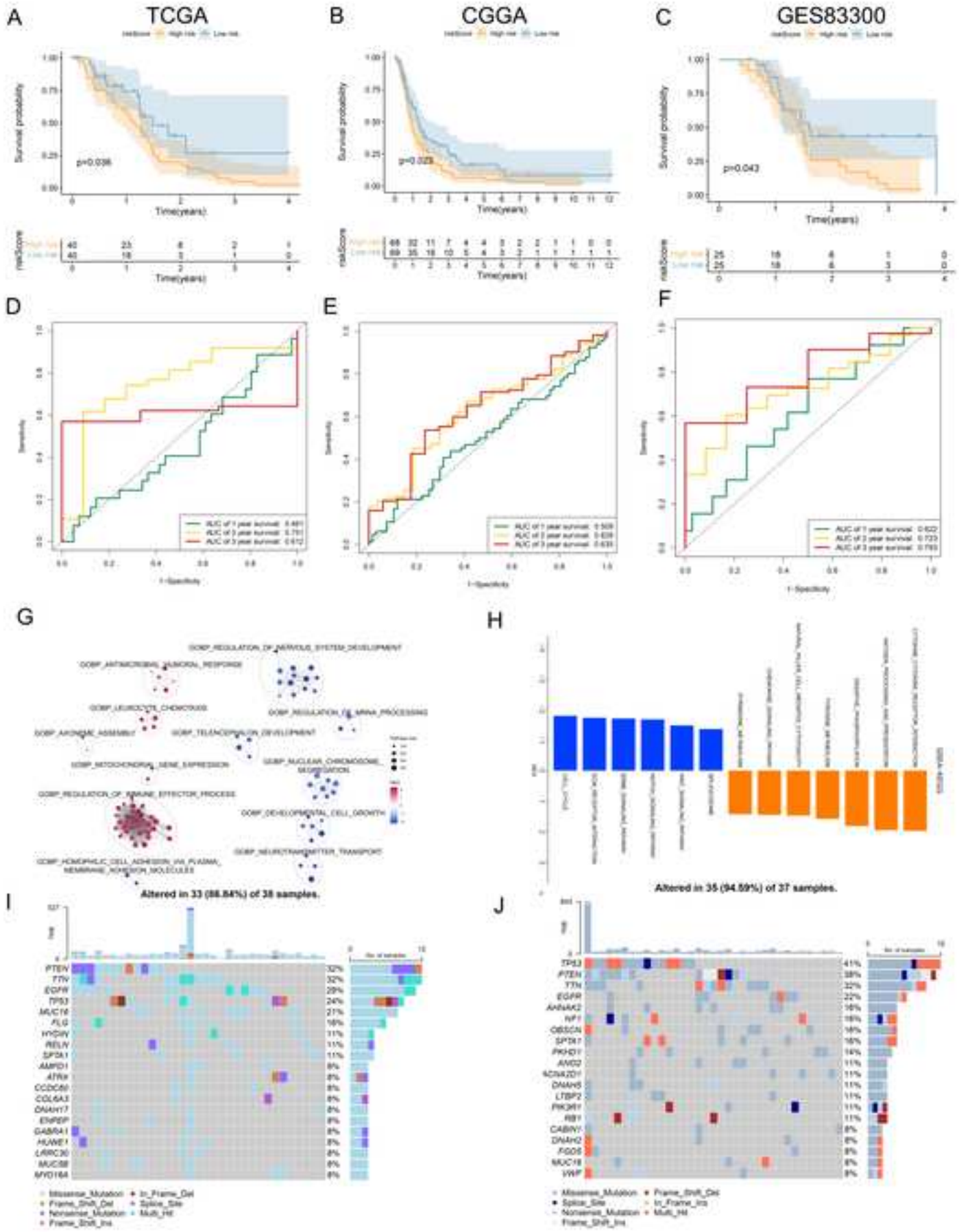

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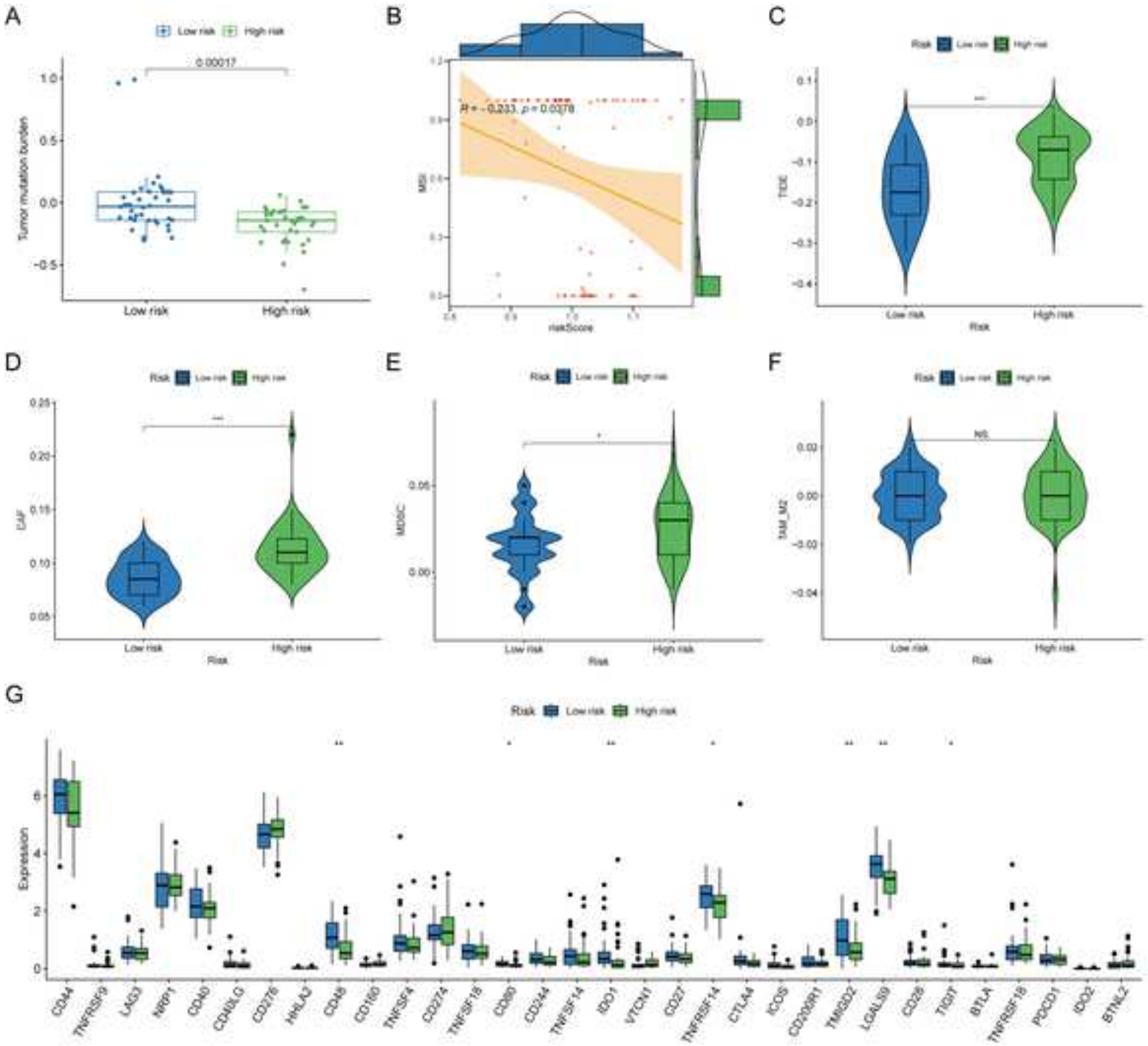

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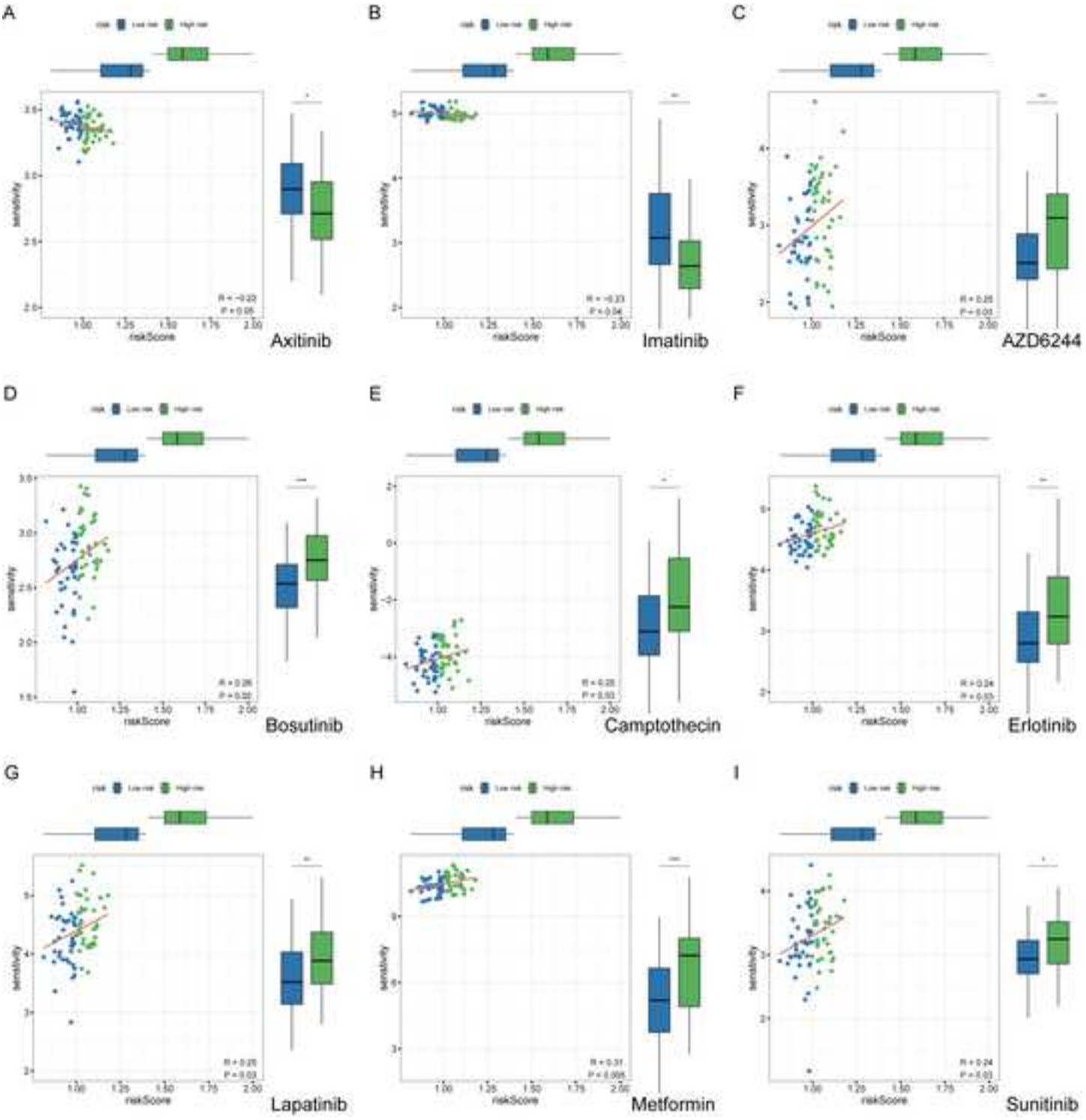

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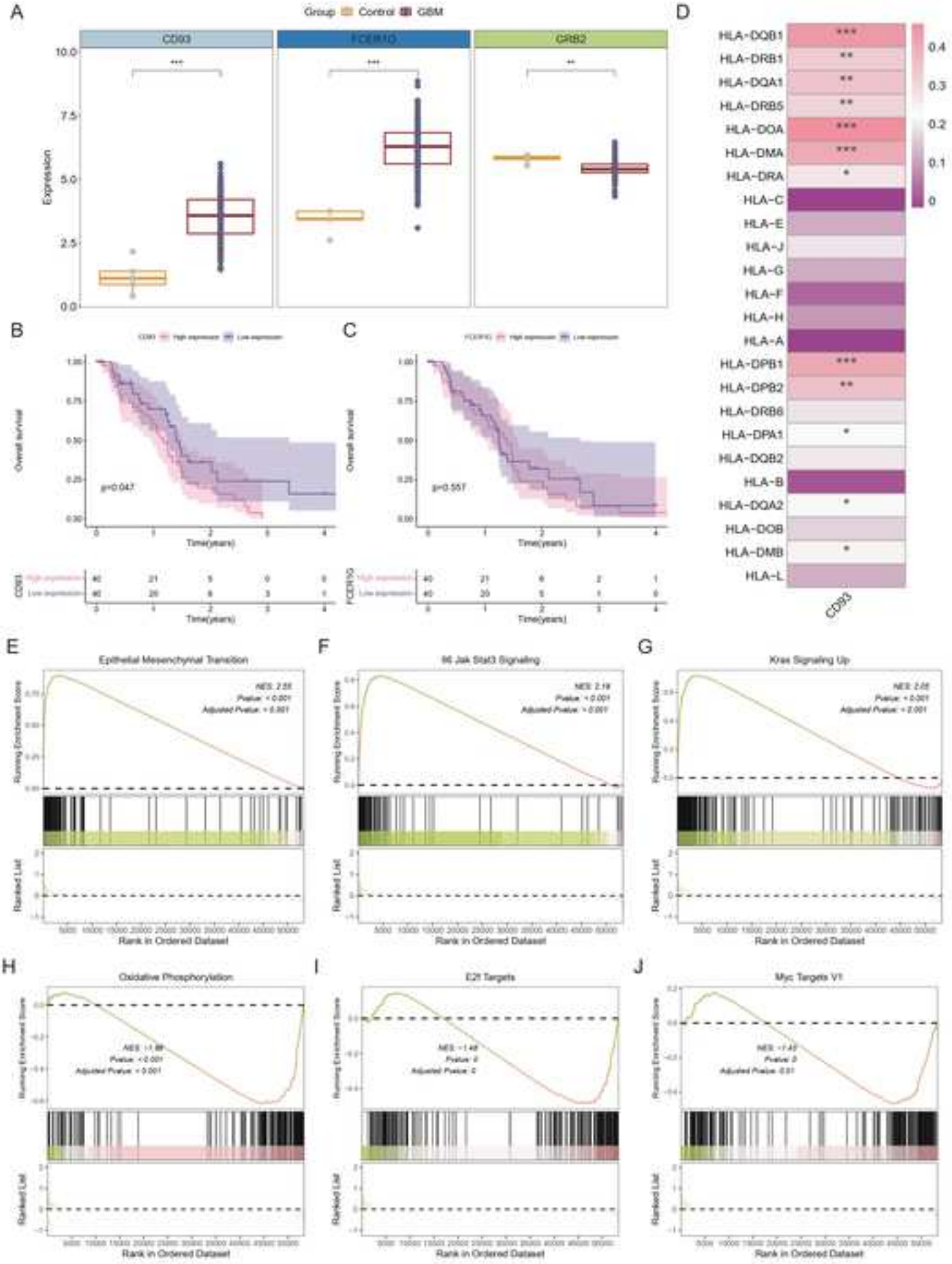

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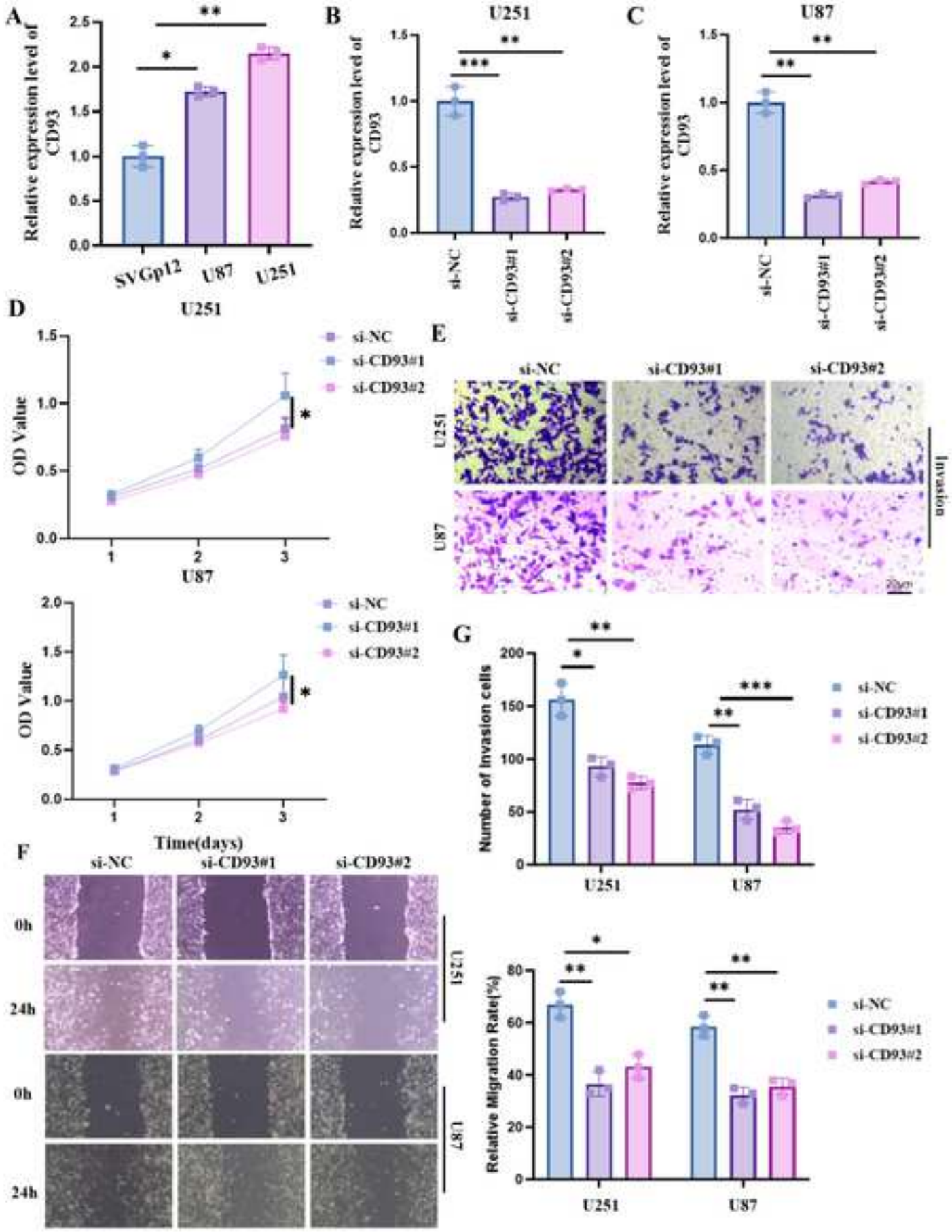

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Supporting Information
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Integrated single-cell RNA sequencing and Bulk-RNA technologies reveal the immunological characteristics of lactylation related-genes in glioblastoma

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Abstract

Objective:

Glioblastoma (GBM) stands as the most aggressive intracranial malignant tumor, characterized by an extremely poor prognosis. Lactylation, a recently identified post-translational modification, has been associated with tumorigenesis. However, its precise role in GBM is not yet fully understood. This study seeks to integrate single-cell RNA sequencing (scRNA-seq) and bulk RNA sequencing (RNA-seq) data to develop a novel prognostic model for GBM, concentrating on lactylation-related factors.

Methods: We have studied lactate metabolism genes as a marker in GBM. We obtained bulk transcriptomic data from TCGA and the GSE141383 and GSE162631 cohorts from GEO databases. We used R package Seurat to analyze scRNA-seq data, cell communication using CellChat R package and AUCell package to assess the lactate metabolism gene set score across cell types. We developed a prognostic model using machine learning algorithms, and tested its efficacy across multiple cohorts. We also investigated differences between high and low risk groups in immune infiltration, predicted sensitivity, and other factors. We validated the function of the key gene CD93 on cell level.

Results: The scRNA-seq data identified nine major cell types in GBM and FCGBP+ macrophages had the highest score of lactate metabolism gene set. Machine learning identified three key genes CD93, FCER1G and GRB2 and designed a model for optimal prognostic value across cohorts. The high-risk group had a high risk outcome. Immunological analysis showed significant differences in the immune cell infiltration and checkpoint gene expression between risk groups. High-risk patients had lower immune infiltration and higher immunosuppression, making them less suitable for immunotherapy. Predictive algorithms suggested that axitinib and imatinib could be potential therapeutic drugs for high-risk patients. In GBM tissue and cells, CD93 expression was significantly increased, thus CD93 was recognized as a key risk gene in this model. Inhibition of CD93 expression by siRNA significantly reduced the proliferation, invasion and migration of U87 and U251 cells.

Conclusion: In summary, we developed a novel characterization of lactylation-related clusters using single-cell sequencing technology. This study offered insights into the prognostic significance of lactate metabolism-related genes in GBM.

Keywords:Glioblastoma, scRNA-seq, prognosis, Machine learning, Immunotherapy.

Introduction

GBM is one of the most common primary intracranial tumors [1-2]. It is the most aggressive tumor of brain neuroglia and is the most commonly encountered primary

intracranial tumor [3-5]. It is rapidly growing and densely infiltrating the brain and hence the deadliest tumor of the central nervous system [3-5]. Initial symptoms include nonspecific cognitive deficits and seizures [5]. Surgery, radiotherapy, and chemotherapy drugs are standard treatments. The median survival rate for GBM patients is under one year [6]. Recent work has identified several biomarkers that provide valuable information about the molecular subtypes of GBM, their prognosis and treatment responses [7]. However, due to its rapid growth and heterogeneous nature, it still remains difficult to treat for oncology. Finding therapeutic targets and developing prognostic risk models are critical for optimal clinical management of glioblastoma.

Since Warburg effect was introduced in 20th century, research has shown lactic acid plays a key role in initiation and progression of malignant tumors [8].Glucose is considered to be the main energy source for tumor cells but tumor cells need a large amount of nutrients and oxygen for rapidly expanding tumor cells [9]. According to Warburg effect, tumor energy metabolism depends more on anaerobic glycolysis than oxidative phosphorylation even in the presence of oxygen. Increased lactic acid levels in tumor microenvironment lead to hypoxic microenvironment [10]. High levels in tumor microenvironment correlate to tumor growth, metastasis, and tumor cell evasion [11]. Tumor cells export large amounts of lactic acid to adjacent immune and stromal cells. Thus, lactic acid acts as mediator between endogenous metabolic processes and immune suppression [12]. She X[13] and colleagues showed that SETDB1 methylation inhibits interaction between MCT1 and Tollip and induces M2-like polarization of tumor-associated macrophages. Su J[14] and colleagues found that absence of lactic acid/GPR81 inhibits expression of chemokine CX3CL1, decreases Treg infiltration into tumor microenvironment (TME) and inhibits gastric cancer progression by improving CD8+ T cells function. The mechanisms of lactic acid production pathways or its transport to lower lactic acid concentrations is not fully understood.

We used scRNA-seq to investigate the role of lactylation in GBM cell diversity. We developed and validated a novel model using signature genes from FCGBP+

macrophages, which showed highest lactylation scores. We also studied the relationship between risk model and immune-infiltrating cells to identify potential therapeutic targets. We identified CD93 as important gene in the model and studied its effects on GBM cell function at cell level.

Methods

Data Collection

The data in this article were sourced from publicly accessible databases, specifically the TCGA (https://portal.gdc.cancer.gov/) and GEO (https://www.ncbi.nlm.nih.gov/geo/). The single-cell datasets used in this study, GSE141383 and GSE162631, include 13 GBM tissue samples and 4 control tissue samples. The bulk datasets are divided into two parts: 1) The training dataset for the diagnostic model comes from the TCGA-GBM cohort, comprising 5 control tissue samples and 168 GBM tissue samples, with survival information for 80 GBM patients; 2) The validation dataset for the diagnostic model is obtained from the CGGA-GBM database, which includes 137 GBM patients with survival information, and the GEO database (GSE83300), which contains 50 GBM patients with matched survival data.

Data Processing

For single-cell data analysis, Seurat version 4.2.2 was employed. Initially, quality control excluded cells with fewer than 200 or more than 5000 nFeature_RNA and those with a mitochondrial proportion over 10%. Data normalization and standardization were conducted using the SCTransform function. Dimensionality reduction and clustering were performed by selecting the top 15 principal components, and the RunUMAP function was used to generate results. The DoubletFinder tool removed doublets from the dataset. To address batch effects across datasets and samples, the Harmony algorithm was applied. Cell annotation was conducted using the SingleR method alongside existing literature. The FindAllMarkers function identified marker genes for each cell type. UMAP results were visualized using the ggplot2 package, and cell proportions were represented with the ggalluvial package.

For bulk RNA data, normalization and standardization were performed using the limma package. Additionally, survival analysis and curve plotting were executed with the Survival and survminer packages.

Ro/e Analysis

The Ro/e ratio [15] illustrates the relationship between observed and expected cell counts for each cell cluster across different tissue types. Expected cell counts for each cell cluster and tissue type combination are derived from chi-square tests. Unlike chi-square values, which only indicate the difference between observed outcomes and random expectations, Ro/e values reveal whether specific cell clusters are enriched or depleted in particular tissues. A Ro/e value greater than 1 suggests that a cell cluster is more prevalent in a specific tissue than expected by chance, indicating enrichment. Conversely, a Ro/e value less than 1 indicates that a cell cluster is less prevalent than expected, suggesting depletion. By employing Role calculations, researchers can effectively evaluate the tissue preferences of various cell clusters.

Cell Communication

The "CellChat" R package (version 1.5.0) is a tool designed to elucidate potential mechanisms of intercellular communication at the single-cell level. The function createCellChat generates a cellchat object, while the aggregateNet feature outlines the signaling outputs for each cell type. The netAnalysis_computeCentrality function calculates the input and output weights of specific signaling pathways. Data visualization is supported by the "CCPlotR" package, where the cc_heatmap function shows the frequency of intercellular communications, and the cc_arrow function illustrates receptor-ligand interactions relevant to target cells.

lactylation geneset score

Gene sets related to lactate were obtained from the MSigDB database (https://www.gsea-msigdb.org/). We calculated the lactylation gene set score using the AUCell package, specifically the AUCell_calcAUC function. This approach enabled

us to compare lactate metabolism gene set scores across various cell types, including macrophage subclusters identified through secondary clustering, to identify cell subpopulations associated with lactate. We visualized the lactate metabolism gene set scores using ggplot2, presenting them as violin plots.

Machine Learning and Prognostic Model Construction

Marker genes linked to the primary lactate-producing cell subtypes are identified using criteria of a log fold change (logFC) greater than 0.5 and a p-value less than 0.001. Core gene identification is performed using three machine learning methods: 1) LASSO regression with the "glmnet" package, 2) the CoxBoost algorithm via the "CoxBoost" package with an initial penalty of 1000, and 3) a Random Forest survival model through the "RandomForestRSF" package, using 1000 trees and evaluating gene significance with importance scores. All methods incorporate cross-validation. A Venn diagram analyzes the intersection of core genes identified by these algorithms to determine core risk prognostic genes. A multivariate regression analysis then develops the prognostic model. Risk scores are calculated by summing gene expression values weighted by their risk coefficients. GBM patients are classified into high-risk and low-risk groups based on the median risk score. Survival analyses and survival curves are generated using the "Survival" and "survminer" packages, while Receiver Operating Characteristic (ROC) curves are created with the "timeROC" package. The "ggplot2" package is used to produce Principal Component Analysis (PCA) plots, aiding in identifying key characteristics of the risk groups and enabling dimensionality reduction clustering.

Enrichment Analysis

Enrichment analysis includes Gene Ontology (GO), Kyoto Encyclopedia of Genes and Genomes (KEGG), and HALLMARK enrichment analyses. The clusterProfiler package is used for Gene Set Enrichment Analysis (GSEA) across these analyses, applying a threshold of an adjusted p-value below 0.05. For visualizing GO enrichment, the aPEAR package is used, while ggplot2 is applied for

188 KEGG enrichment visualization. Additionally, the GseaVis package is utilized to
189 visualize HALLMARK enrichment analysis.
190
191 **Immunology-related Analysis**
192 ssGSEA is a widely used method for evaluating immune cell types, focusing on
193 29 distinct categories and related immune processes. The "GSVA" and "GSEABase"
194 packages calculate relative scores for these immune cells and processes. For
195 visualization, the "ggpubr" and "introdataviz" packages create split violin plots. The
196 "Estimate" package analyzes the tumor microenvironment, and correlation analyses
197 explore relationships between risk scores and immune, stromal, and
198 microenvironment scores. The "TCGAbiolinks" package retrieves single nucleotide
199 polymorphism (SNP) data for GBM patients, while "maftools" calculates tumor
200 mutational burden (TMB) scores for each patient. Additionally, the TIDE website
201 (http://tide.dfci.harvard.edu/) assesses immune evasion and dysfunction status. Finally,
202 "ggpubr" generates boxplots to depict the expression levels of immune checkpoint
203 molecules.
204 **Drug Sensitivity Analysis**
205 The pRRophetic algorithm was used to calculate drug sensitivity IC50 values,
206 aiding in predicting patient responses to chemotherapeutic and targeted agents across
207 different cohorts. By analyzing the correlation between risk scores and drug IC50
208 values, drugs with positive correlation coefficients emerged as potential therapeutic
209 options for low-risk groups. In contrast, drugs showing a significant negative
210 correlation with IC50 values were identified as potentially suitable for high-risk
211 patients.
212
213 **Cell Culture**
214 Human brain astrocytes SVG p12 (FH0563) and glioblastoma cell lines U251
215 (FH0159) and U87 (FH0162) were obtained from Fuheng Biological Technology Co.,
216 Ltd. (Shanghai, China). These cells were cultured in DMEM medium (Gibco, Grand
217 Island, NY, USA) supplemented with 10% fetal bovine serum (FBS; Gibco), 100

218 U/mL penicillin, and 100 μg/mL streptomycin (Invitrogen, Waltham, MA, USA).
219 Cultivation occurred in a humidified incubator set at 37°C with 5% CO2.
220
221 **Cell Transfection**
222 The siRNA targeting CD93 was designed and synthesized by Tsingke
223 Biotechnology Co., Ltd. (Beijing, China). Transfection was performed using
224 Lipofectamine 3000 reagent (Invitrogen, Waltham, MA, USA) according to the
225 manufacturer's instructions. Post-transfection, cells were cultured at 37°C with 5%
226 CO2 for 48 hours. The study comprised three experimental groups: cells transfected
227 with si-CD93#1, si-CD93#2, and a negative control group (si-NC). The specific
228 siRNA sequences used in the study were as follows:
229 si-CD93#1:GGCUACUGGUCUAUCGCAA;
230 si-CD93#2: GUGCAAGUUCAGCUUCAAA;
231
232 **Cell Viability Assay**
233 Cell viability was assessed using the CCK-8 assay kit (Invigentech, IV08, USA).
234 After 48 hours of siRNA transfection, U251 and U87 cell lines were seeded into
235 96-well plates at a density of 3,000 cells per well. At 24, 48, and 72 hours
236 post-transfection, 10 μL of CCK-8 reagent was added to each well, followed by a
237 2-hour incubation at 37°C. The optical density (OD) at 450 nm was then measured
238 with a microplate reader to determine the relative cell survival rate.
239
240 **Transwell Invasion Assay**
241 The invasion assay employed Transwell chambers with 8μm pores (Corning,
242 USA, Cat. No. 3422) placed in a 24-well plate. The upper chamber was coated with
243 an 8-fold dilution of BD Matrigel. U251 and U87 cell lines, transfected with siRNA
244 for 48 hours, were digested with trypsin and resuspended at 3×104^5 cells per
245 chamber in 200μL of serum-free medium. These cells were added to the upper
246 chamber, while 600μL of medium containing 20% fetal bovine serum (FBS) was
247 placed in the lower chamber. The setup was incubated at 37°C with 5% CO2 for 48

248 hours. After incubation, non-migratory cells were removed from the upper chamber,
249 and cells adhering to the membrane underside were fixed with 4% paraformaldehyde
250 for 20 minutes. Following fixation, the cells were stained with crystal violet. Cell
251 invasion was observed and documented using a microscope..
252
253 **Scratch assay**
254 Forty-eight hours after siRNA transfection, U251 and U87 cell lines were
255 re-plated into 6-well plates and cultured until they reached about 90% confluence
256 within 24 hours. Using sterile 10 μl pipette tips, three linear scratches were made on
257 the cell monolayer. The wells were then rinsed with phosphate-buffered saline (PBS)
258 to remove any detached cells. A medium containing 4% fetal bovine serum (FBS) was
259 added to each well, and the cells were further cultured. The wound healing process
260 was observed, and images of the scratched regions were captured at 0 and 24 hours
261 using a microscope.
262
263 **Quantitative Real-Time PCR (qRT-PCR)**
264 U251 and U87 cells were collected 48 hours post-siRNA transfection. The
265 procedure followed the kit's instructions. Total RNA was extracted using the RNAiso
266 Plus Reagent Kit (Takara, 9108/9109, Japan). mRNA was then reverse-transcribed
267 into cDNA with the PrimeScript RT Reagent Kit (Takara, RR037A, Japan). Real-time
268 quantitative PCR was conducted using the TB Green Premix Ex Taq II Reagent Kit
269 (Takara, RR820A, Japan). GAPDH served as the normalization control, and CD93
270 mRNA expression levels were determined using the 2-ΔΔCt method. The primer
271 sequences are as follows:
272 CD93-F: 5'-GGCAGACAGTTACTCCTGGGTT-3',
273 CD93-R: 5'-GGAGTTCAAAGCTCTGAGGATGG-3';
274 GAPDH-F：5’- ACAGCCTCAAGATCATCAGCA-3’;
275 GAPHD-R: 5’-ATGAGTCCTTCCACGATACCA-3’;
276
277 **Data Analysis**

Statistical analyses utilized R software (version 4.3.1). Continuous variables were analyzed with either Student's t-test or the Mann-Whitney U test, while categorical variables were assessed using the chi-square test or Fisher's exact test. Kaplan-Meier survival analysis estimated survival outcomes. For analyses with multiple comparisons, such as differences in clinical features across risk groups, P-values were adjusted for the false discovery rate using the Benjamini-Hochberg method. Statistical significance was denoted as follows: *P < 0.05, **P < 0.01, ***P < 0.001, and NS for no significant difference.

3 Result

The scRNA profiling of GBM

To thoroughly understand the cell types and molecular changes linked to GBM, we integrated two single-cell datasets (GSE141383 and GSE162631) and used the Harmony algorithm to correct for batch effects. After implementing stringent control measures, we analyzed 114,630 cells, including 45,282 control cells and 69,348 primary GBM cells. Using a resolution parameter, we categorized these cells into nine distinct populations (Fig. 1A). Specific annotations were based on markers for macrophages (OC1, SPP1, CD13), microglia (CX3CR1, P2RY12, P2RY13), neuron-like cells (PTN, PTPRZ, BCAN), monocytes (VCAN, LYZ, THBS1), neutrophils (IL12, CSF3R), T cells (CD2, CD3D, CD3E), endothelial cells (CLDN5, VWF), fibroblasts (RGS5, DCN, COLA1), and proliferating cells (MKI67, TOP2A, TPX2) (Fig. 1B).To identify cell types closely associated with GBM progression, we conducted an observed-to-expected ratio (Ro/e) analysis, a method commonly used to assess tissue preference. Results indicated that microglia and endothelial cells were predominantly enriched in control tissues, while other cell types were concentrated in GBM tissues (Fig. 1C). This finding suggests that tumorigenesis mobilizes various cells to collaboratively reshape the tumor microenvironment. We further evaluated the relative differences in cell types between GBM and control tissues. Notably, macrophages, neuron-like cells, monocytes, and neutrophils significantly increased in

GBM, highlighting their critical roles in GBM development and progression (Fig. 1D).Cell communication analysis revealed that macrophages and microglia exhibited the highest incoming and outgoing signal intensities (Fig. 1E). Although increasing evidence suggests that excessive lactylation activation is closely associated with tumor progression, its impact on GBM progression remains unclear. We obtained a lactylation-related gene set from the MSigDB database and applied the AUCell score to evaluate the nine cell types. Macrophages had the highest lactylation gene set score (Fig. 1F), leading us to speculate that macrophages may promote GBM progression through the lactylation pathway.Subsequently, we performed a secondary clustering analysis of macrophages and identified 11 different macrophage subtypes (Fig. 1G). Interestingly, the FCGBP+ macrophage subtype showed the highest lactylation gene set score (Fig. 1H). Additionally, we compared lactylation gene set scores between the control and GBM groups, finding that the GBM group had a significantly higher score ($P < 0.01$), further confirming lactylation as a important factor in GBM occurrence and development (Fig. 1I).

Machine Learning Construct FCGBP+ Macrophages Signatures

We identified marker genes of FCGBP+ macrophages using screening criteria (logFC > 0.5, P 0.001) and validated their expression in control and GBM tissues of TCGA-GBM cohort. We found 12 genes upregulated and 9 downregulated in GBM (Fig. 2A). GO enrichment analyses of these differentially expressed genes showed their role in immune cell regulation, cell differentiation, and other biological processes (Fig. 2B). To identify core genes of FCGBP+ macrophages, we applied three machine learning algorithms. CoxBoost algorithm obtained the most stable model at the 117th boost and identified 3 core genes with partial log log 210.4623 (Fig. S1A); LASSO regression allowed shrinkage and cross validation and retained 3 most reliable core genes lambda.min = 0.1060313 (Fig. S1B-C). Random forest survival model obtained the lowest error rate of 0.4783 at 143 trees, ranking differential genes by importance and identified 4 genes with an importance score above zero (Fig. S1D-E).We constructed Venn diagram from outputs of the three

337 machine learning algorithms and selected intersecting genes CD93, FCER1G, and
338 GRB2 as core genes (Fig. 2C).
339
340 **Validation of Prognostic Features and Clinical Value**
341 We developed a risk model based on three lactylation genes: CD93, FCER1G,
342 and GRB2. We calculated the risk score for each GBM patient using Lasso analysis
343 and multivariate Cox regression analysis. Patients were then classified into high-risk
344 and low-risk groups based on the median risk score. In the training cohort
345 (TCGA-GBM dataset) and two validation sets (CGGA dataset, GSE83300), we
346 observed a significant difference in overall survival between these groups. High-risk
347 patients exhibited lower survival rates compared to low-risk patients ($P < 0.05$) (Fig.
348 3A-C).To evaluate the predictive power of these three genes for GBM prognosis, we
349 conducted time-dependent receiver operating characteristic (ROC) curve analyses in
350 both training and validation sets. The average area under the curve (AUC) of the risk
351 model at 1, 2, and 3 years was 0.55, highlighting its biological significance (Fig.
352 3D-F). We also identified significant biological processes and pathways
353 differentiating low-risk and high-risk patients. Gene Ontology (GO) enrichment
354 analysis revealed that low-risk patients had higher immune effector processes and
355 increased activation and expression of mitochondrial genes (Fig. 3G).Further, KEGG
356 enrichment analysis showed that pathways related to cytokine receptor interaction,
357 antigen presentation, and oxidative phosphorylation were upregulated in the low-risk
358 group, while cancer pathways such as WNT and NOTCH were downregulated (Fig.
359 3H). These findings help elucidate the reasons for the differences observed between
360 the risk groups. The low-risk group had a higher mutation frequency (94.59%), with
361 TP53 being the most commonly mutated gene (Fig. 3I-J). In contrast, the high-risk
362 group had a mutation frequency of 86.84%, with mutations primarily in the PTEN
363 gene (Fig. 3I-J).
364
365 **Immune-Related Features of the Risk Model**
366 We assessed immune infiltration across various risk categories and core genes

using the ssGSEA algorithm to calculate scores for 29 distinct immune cell types and processes. Our findings indicated that the low-risk group had higher infiltration levels of dendritic cells (DCs), neutrophils, T helper cells, and tumor-infiltrating lymphocytes (TILs). This group also exhibited enhanced T cell costimulation and type I interferon responses (Supplementary Fig. S2A). Additionally, the core genes associated with the three models showed unique relationships with TNF family genes and chemokines. Notably, FCER1G had significant negative correlations with most TNF family genes and chemokines, while CD93 and GRB2 showed partial positive correlations with these genes and chemokines (Supplementary Fig. S2B-C). To evaluate TME activity, we used TME scores and discovered that high-risk scores were significantly negatively correlated with both immune scores and TME scores. This suggests that lower risk scores may be associated with more immunologically active microenvironments and relatively lower tumor purity (Supplementary Fig. S2D-F).

Immunotherapy Efficacy Stratification by Immune Microenvironment Risk Model

We evaluated the effectiveness of immunotherapy across different risk categories and discovered that the tumor mutation burden (TMB) was higher in the low-risk cohort. Furthermore, we found a significant negative correlation between microsatellite instability (MSI) and risk scores, indicating that low-risk patients have elevated levels of both TMB and MSI, making them more suitable for immunotherapy (Fig. 4A-B). Conversely, the high-risk group showed significantly increased TIDE scores and a higher prevalence of tumor-associated fibroblasts and myeloid-derived suppressor cells (MDSCs), suggesting a connection between high-risk status and less favorable immunotherapy outcomes (Fig. 4C-F). Our analysis also revealed that immune checkpoint molecules were predominantly expressed at higher levels in the low-risk group, reinforcing the notion that these patients are more likely to benefit from immunotherapy (Fig. 4G). Therefore, individuals in the low-risk group may represent a promising demographic for immunotherapeutic strategies.

397 **Analysis of Antitumor Drugs in High-Risk and Low-Risk Groups**

398 We conducted a drug sensitivity prediction analysis for patients across various
399 risk groups, utilizing the pRRophetic algorithm to determine IC50 values. Axitinib
400 and Imatinib displayed a significant negative correlation with the risk score,
401 indicating their potential effectiveness for high-risk patients (Fig. 5A-B). Conversely,
402 seven drugs, including AZD6244, showed a positive correlation with the risk score,
403 suggesting their appropriateness for low-risk patients. Notably, metformin exhibited
404 the strongest and statistically significant correlation (Fig. 5C-I).

405

406 **CD93 is potentially a important gene in models linked to poor prognosis in GBM**

407 To identify key molecular biomarkers for our prognostic model, we evaluated the
408 expression levels of three critical genes in GBM. Our analysis revealed that only
409 CD93 and FCER1G were significantly upregulated in GBM tissues, suggesting their
410 potential roles in tumor progression (Fig. 6A). Kaplan-Meier survival analysis further
411 indicated that CD93 was the only gene with a statistically significant impact ($P<0.05$)
412 on GBM patient survival, with higher CD93 expression associated with poorer overall
413 survival outcomes (Fig. 6B-C). Therefore, CD93 emerged as a promising molecular
414 biomarker for GBM.Interestingly, CD93 did not significantly affect the tumor antigen
415 presentation process but showed a positive correlation with HLA family molecules
416 (Fig. 6D), implying involvement in alternative tumor-promoting mechanisms.
417 Additionally, hallmark enrichment analysis compared pathway variations between
418 groups with high and low CD93 expression. Patients with elevated CD93 expression
419 demonstrated increased activity in the epithelial-mesenchymal transition (EMT),
420 interleukin-6 (IL6), and KRAS pathways. In contrast, those with lower CD93
421 expression exhibited upregulation of processes such as oxidative phosphorylation (Fig.
422 6E-J). These pathways may elucidate how CD93 contributes to GBM progression.

423

424 **CD93 is upregulated in GBM cell lines and enhances GBM cell proliferation,**
425 **invasion, and migration**

426 In summary, the findings indicate that CD93 is a significant gene associated with

427 poor prognosis in GBM. Quantitative reverse transcription polymerase chain reaction
428 (qRT-PCR) analyses showed that human astrocytoma cells had higher CD93
429 expression levels compared to human brain astrocytes (Fig. 7A). We performed
430 simultaneous transfections of siRNA#1 and #2 into U251 and U87 cell lines, which
431 confirmed the transfection efficiency of CD93 (Fig. 7B-C). Cell Counting Kit-8
432 (CCK8) assays demonstrated that knocking down CD93 significantly inhibited the
433 proliferation of both U251 and U87 cells (Fig. 7D). Additionally, scratch and
434 Transwell assays revealed that CD93 knockdown decreased the migratory and
435 invasive abilities of U251 and U87 cells. Overall, these results suggest that CD93 acts
436 as a positive regulator in the context of GBM.
437

438 ## Discussion

439 This study innovatively integrates single-cell sequencing data with extensive
440 transcriptome datasets from public databases to investigate the lactic acid metabolism
441 mechanism in macrophages within the GBM tumor microenvironment. We performed
442 a comprehensive analysis of GBM by incorporating large-scale transcriptome data.
443 Utilizing single-cell sequencing, machine learning, and immune infiltration analysis,
444 we examined the effects of lactation modification on GBM. Our findings suggest that
445 genes associated with lactation modification could serve as novel therapeutic targets.
446 Additionally, we established a reliable patient survival prognosis model.The
447 importance of lactation modification in the glioblastoma tumor microenvironment is
448 increasingly acknowledged, with lactate dehydrogenase (LDH) identified as a major
449 prognostic marker for this malignant tumor [16]. Tumor cells often undergo metabolic
450 reprogramming, resulting in excessive lactic acid production, which promotes
451 angiogenesis, immune evasion, and drug resistance [17-18]. Research by Yan et al.
452 [19] confirmed that lactic acid enhances GPR65 expression on tumor-associated
453 macrophages (TAMs), activating the cAMP/PKA/CREB signaling pathway, thereby
454 facilitating glioblastoma invasion and metastasis. Consequently, lactic acid
455 modification presents a promising target for GBM treatment.Further research is

456 essential to pinpoint specific targets related to lacylation, which could become new
457 therapeutic targets for GBM. Developing innovative therapeutic strategies based on
458 this metabolic pathway is prominent.

459 We obtained RNA expression profile data for GBM from the TCGA database,
460 while gene mutation data and clinical information came from the external validation
461 cohort in the GEO and CGGA databases. This comprehensive dataset enabled a
462 thorough analysis of GBM's molecular characteristics. Additionally, we used
463 single-cell RNA sequencing data to explore GBM heterogeneity within the tumor
464 microenvironment, identifying nine distinct cell types: macrophages, microglia,
465 neuron-like cells, monocytes, neutrophils, T cells, endothelial cells, fibroblasts, and
466 proliferation-like cells. These findings provide meaningful insights into the
467 mechanisms underlying GBM occurrence.We employed the lactic acid metabolism
468 gene set score to identify individual cells involved in lactic acid metabolism.
469 Macrophages exhibited the highest gene set score, aligning with Yang et al.'s findings
470 [20], which reported a significant link between lactic acid metabolism and gastric
471 cancer progression, with macrophages showing the highest infiltration levels. This
472 supports our results. To pinpoint the primary cellular subtypes influencing lactic acid
473 metabolism, we conducted secondary dimensionality reduction clustering, classifying
474 macrophages into 11 subtypes. The FCGBP+ macrophage subtype displayed the
475 highest lactate metabolism gene set score.Subsequently, we analyzed core genes
476 related to the FCGBP+ macrophage subtype in the TCGA-GBM cohort, identifying
477 12 up-regulated and 9 down-regulated genes in GBM. These differentially expressed
478 genes are involved in immune cell regulation, cell differentiation, and various
479 biological pathways. Our research deepens the understanding of lactic acid metabolic
480 activity variations in GBM cells.The primary objective of this study was to establish a
481 robust prognostic model for GBM. Using machine learning algorithms, we
482 constructed a predictive model based on key lactic acid metabolism-related genes.
483 This model effectively stratifies patients into high-risk and low-risk groups. Survival
484 analysis revealed that high-risk patients have shorter prognoses and survival times.
485 Notably, the risk model, validated across the TCGA training set, CGGA validation set,

and GEO validation set, achieved an average area under the curve (AUC) of over 0.6 at 1, 2, and 3 years. This demonstrates the model's stability and accuracy in multi-cohort studies and its potential clinical application.Furthermore, our study conducted an in-depth analysis of the immune characteristics of GBM, revealing significant differences between high-risk and low-risk groups in immune cell infiltration, immune checkpoint gene expression, and tumor mutation burden. These findings suggest that the prognostic model's risk stratification mechanism may reflect immunological differences among GBM tumors in various populations, highlighting the need to explore differential immunotherapy responses among patient groups.

Through the analysis of core lactate-promoting genes, we identified CD93 as a key gene linked to poor prognosis in GBM. Polymeric protein-2 (MMRN2), a widely expressed extracellular matrix protein, specifically interacts with CD93, promoting endothelial cell adhesion and migration, which facilitates angiogenesis [21]. Additionally, CD93 is significantly upregulated in breast cancer, where its interaction with MMRN2 activates integrin β1, promoting tumor growth and angiogenesis via the PI3K/AKT/SP2 signaling pathway [22]. Thus, we speculate that CD93 could serve as a potential biomarker for GBM. Cell experiments confirmed that high CD93 expression in GBM cells, and inhibiting CD93 expression significantly reduced the proliferation, invasion, and migration abilities of U87 and U251 cells. This finding offers promising therapeutic targets for future GBM treatments.In conclusion, this study demonstrated that using the lactic acid fermentation gene set score as an indicator in single-cell datasets effectively identifies active cell clusters involved in lactic acid metabolism. Research found that this specific cell cluster is associated with a more aggressive disease phenotype, suggesting its important role in GBM progression. By integrating single-cell RNA sequencing data with bioinformatics technology, we identified key genes related to lactic acid fermentation and developed an innovative diagnostic model for predicting GBM. These results reveal lactic acid-mediated metabolic reprogramming and immune escape mechanisms in GBM, while identifying potential biomarkers and therapeutic targets. Further validation through larger patient cohorts is necessary, and the potential efficacy of targeting

lactate metabolic pathways in GBM treatment should be evaluated.

GBM:Glioblastoma;

TNF:Tumor Necrosis Factor;

TME:Tumor Microenvironment;

TAM:Tumor-Associated Macrophage;

CAF:Cancer-Associated Fibroblast;

EC:Endothelial Cell;

scRNA-seq:Single-cell RNA Sequencing;

GEO:Gene Expression Omnibus database;

TCGA:The Cancer Genome Atlas database;

AUC: Area Under the Curve

ROC： Receiver Operating Characteristic

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None

Competing interests

The authors declare that there are no competing interests.

Author Contributions

Biao Wang: experimental work, Conceptualization, Investigation, Writing–original draft. **Yangfang An**: Formal analysis. **Yuansen Shu**: Formal analysis,Writing–original draft. **Xiaoping Cheng**: Writing–review & editing.**Xuxiang Chen**: Writing – review & editing. **Xiezhuo Zhang**: Supervision, Project administration, Methodology. **Zhaorui Cheng**: Supervision, Project administration, Methodology.

Data availability statement

The data that support the findings of this study are openly available in Gene Expression Omnibus (GEO) at https://www.ncbi.nlm.nih.gov/geo/, reference number GSE141383，GSE162631.

Ethics aproval and consent to participate

This article does not contain any studies with human participants or animals.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

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Figures legends

Fig.1 Immune landscape of single-cell data in GBM

(A) Identification of nine cell types in GBM. (B) Gene annotation for these nine cell types. (C) Distribution comparison of the nine cell types between control and GBM tissues. (D) R o/e scores for each of the nine cell types. (E) Heatmap illustrating communication among the nine cell types. (F) Lactylation gene set scores for the nine cell types. (G) Comparison of lactylation gene set scores between normal and GBM tissues. (C) UMAP plot depicting 11 cell subtypes following secondary clustering of macrophages. (E) Presentation of lactylation gene set scores between control and GBM tissues. (F) Lactylation gene set scores across 11 macrophage subtypes. (G) Identification of 11 macrophage subtypes in GBM. (H) Comparison of lactylation gene set scores between normal and GBM tissues. (I) Lactylation gene set scores among the 11 macrophage subtypes.

Fig.2 Construction of FCGBP+ Macrophages Prognostic Genes

(A) The volcano plot illustrates the differentially expressed genes in FCGBP+ macrophages. (B) The GO enrichment analysis highlights the differentially expressed genes in FCGBP+ macrophages. (C) The Venn diagram shows the gene intersections identified by three distinct machine learning algorithms.

Fig.3 Validation and enrichment analysis of the risk prognostic model

(A-C) Kaplan-Meier survival curves illustrate overall survival (OS) for patients across various risk models within the TCGA, CGGA, and GSE83300 cohorts. (D-F) ROC curves depict 1-year, 2-year, and 3-year survival rates for different risk models in the

TCGA, CGGA, and GEO GSE83300 datasets. (G) GSEA pathway analysis compares the high-risk and low-risk groups. (H) KEGG pathway analysis contrasts the high-risk and low-risk groups. (I-J) mutational landscape plot provide a summary of mutation rates in high-risk versus low-risk populations. (*p<0.05; **p<0.01; ***p<0.001; NS: not statistically significant)

Fig.4 Characteristics of Multiple Immune Microenvironment Scoring Systems in Risk Models

(A)Analysis of Immune Mutation Burden in High-Risk and Low-Risk Groups (B)Correlation Analysis of MSI Scores Between High-Risk and Low-Risk Groups.(C)TIDE Score Analysis in High-Risk and Low-Risk Groups (D) CAF Score Analysis in High-Risk and Low-Risk Groups (E) MDSC Score Analysis in High-Risk and Low-Risk Groups.(F)TAM_M2 Score Analysis in High-Risk and Low-Risk Groups.(G)Correlation Analysis of Immune Checkpoint-Related Molecules Between High-Risk and Low-Risk Groups.*p<0.05; ** p<0.01; *** p<0.001;NS: not statistically significant.

Fig.5 Predicted drug sensitivity stratified by immune microenvironment risk score

（A–I）The pRRophetic algorithm predicts correlations between nine clinically relevant antitumor drugs and drug sensitivity in patients with varying risk scores.(A) Axitinib, (B) Imatinib, (C) AZD6244, (D) Bosutinib, (E) Camptothecin, (F) Erlotinib, (G) Lapatinib, (H) Metformin, (I) Sunitinib. Box plots compare the distribution of predicted sensitivity scores between low-risk and high-risk groups, where *p<0.05; ** p<0.01; *** p<0.001;NS: not statistically significant.

Fig.6 Prognostic model core gene screening

(A) Expression levels of CD93, FCER1G, and GRB2 in the TCGA-GBM cohort. (B-C) Survival curve analysis of CD93 and CER1G.(D) Correlation analysis of CD93 with human leukocyte antigen (HLA) family genes.(E-J)KEGG pathway enrichment

analysis of CD93-related genes. (*p<0.05; ** p<0.01; *** p<0.001;NS: not statistically significant).

Fig.7 Validation of the role of core gene CD93 in cytology experiments

(A) Expression levels of CD93 mRNA in GBM cell lines. (B-C) Knockdown efficiency of CD93 mRNA in U251 and U87 cells. (D) CCK8 analysis showed that interfering with CD93 expression significantly inhibited the proliferation ability of U251 and U87 cells.(E-F) Interfering with CD93 expression significantly inhibited the migration and invasion abilities of U251 and U87 cells. (G) Statistical graphs of invasion and migration experiments (*p<0.05, **p<0.01, ***p<0.001, ns: not significant).

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Dear. Editor
Thank you very much for giving us a chance to revise our manuscript. The reviewers and editors comments are valuable and very helpful for improving our research paper.

Reviewer #1:
Line 46: "predicted sensitivity" may be prefered here.
Response:
We appreciate the reviewer's careful suggestion. We have revised the relevant text at Line 46, replacing the original expression with "predicted sensitivity" as recommended. This revision makes the description more accurate and consistent with the research context.

- Line 52: Without an external *clinical* confirmation in large scale cohorts, "optimized" might be a preferable wording over "optimal", highlighting the process without being able to claim optimal performance in real world situations.
Response:
We appreciate the reviewer's careful and constructive comment. Following the suggestion, we have revised the word "optimal" to "optimized" at Line 52. This change better emphasizes the model construction process rather than overstating its real-world performance without large-scale external clinical validation.

- Data processing, Line 130 and below: The authors claim that they filtered out cells with >10% mtRNA. However, in their response letter, they state 20%. Please check for accuracy. I also recommend stating the percentage of total variance captured by these 15 PCAs in the manuscript body and not only in the response letter.
Response:
We greatly appreciate the reviewer for catching this critical inconsistency and for the constructive suggestions.We have carefully checked and corrected the mitochondrial RNA filtering threshold in the manuscript (Line 130 and below). The threshold used in our analysis was 10% mtRNA, and we have revised the response letter accordingly to ensure full consistency across the manuscript and response materials.Following the reviewer's recommendation, we have added the percentage of total variance captured by the 15 principal components into the main text of the manuscript, so that this key information is clearly reported and visible to readers.All corrections have been made in the revised manuscript.

- Ro/e analysis, Line 146 and below: In this paragraph, the authors spell "Role" and not "Ro/e" multiple times, which is somewhat confusing.
Response:
We thank the reviewer for pointing out this typographical error.We have carefully checked the paragraph at Line 146 and below, and corrected all misspelled "Role" back to "Ro/e" to avoid confusion and ensure consistency in the manuscript.

- Line 257: Please make sure that all reagents are denoted with their catalogue number and manufacturer. This can either be done in brackets after the first mention of a reagent or in a supplementary reagent table.

Response:
We appreciate the reviewers' constructive suggestions.
We thoroughly reviewed all the reagents listed in the manuscript and included the respective manufacturers in parentheses. The entire manuscript was meticulously checked multiple times to ensure that the source information for all reagents is clearly indicated.

- Line 334: The line still somewhat overstates the results of this study, "crucial". The same occurs in line 485.
Response:
We appreciate the reviewers' detailed comments. We agree that using "crucial" in Lines 334 and 485 overstates our study's conclusions. Consequently, we revised these sentences, replacing "crucial" with more suitable and moderate terms to accurately reflect our findings.

- Discussion: Multiple parapgraphs in the discussion are doubled, as if copied and pasted multiple times. This suggests that the authors have not proof-read the final manuscript. Please make sure that the revised manuscript is free of apparent mistakes. E.g., lines 463-466 & 476; 486 - 517 in multiple snippets.
Response:
Dear Reviewers,We sincerely apologize for the repeated paragraphs and inadvertent errors in the discussion section. These issues resulted from inadequate proofreading prior to submission.
We meticulously revised the discussion section, eliminating redundant content, including the repetitive paragraphs in lines 463–466, line 476, and lines 486–517. We reorganized the paragraphs to maintain logical consistency and thoroughly proofread the manuscript to correct any noticeable errors. The revised manuscript was carefully reviewed to improve its readability and academic quality.